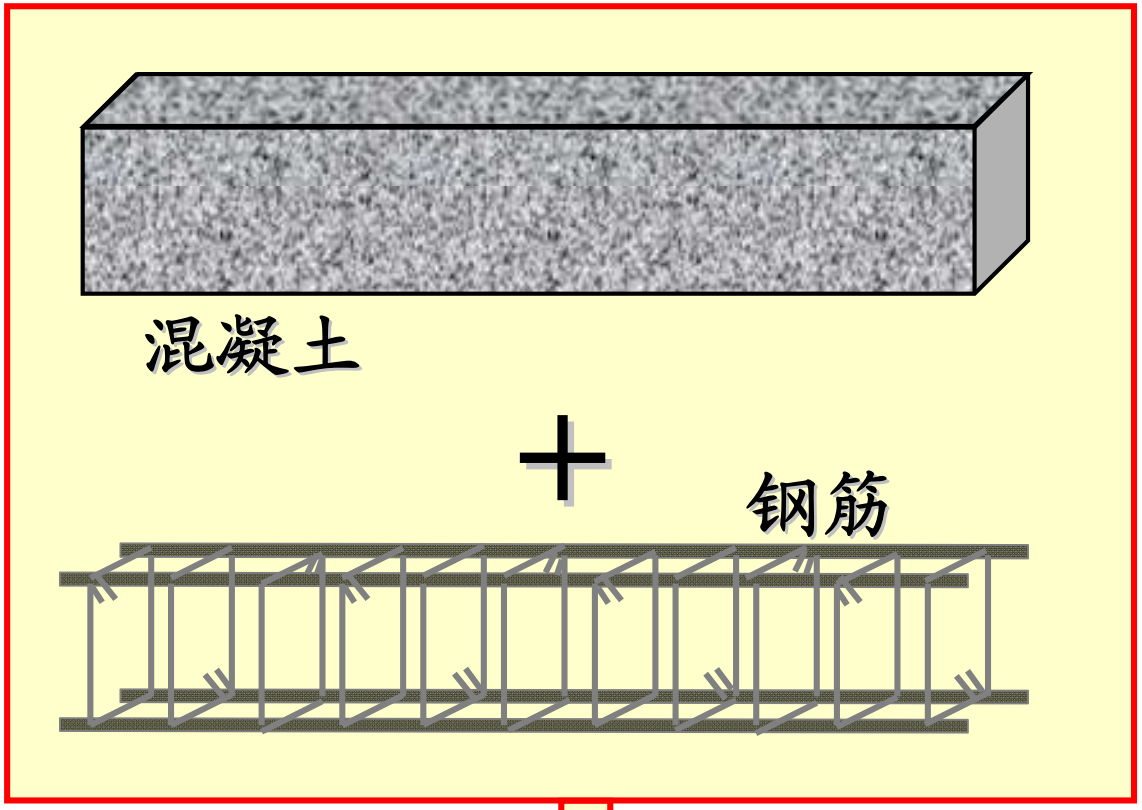
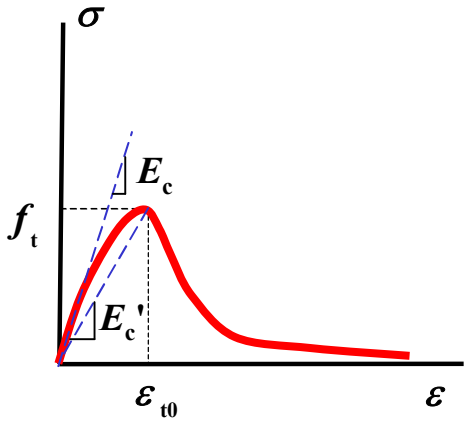
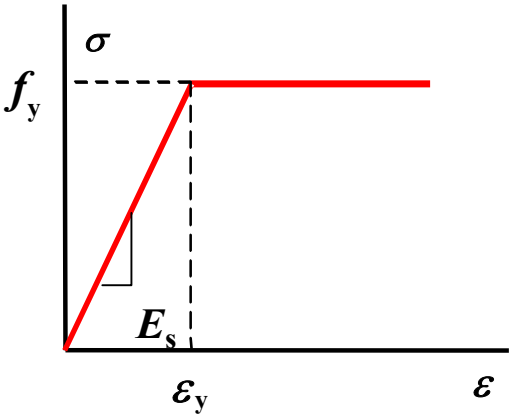
《混凝土结构》

第三章 钢筋混凝土构件的基本受力性能

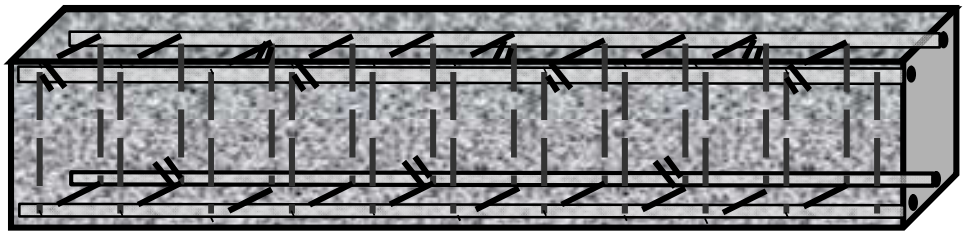
Mechanical Behaviors of RC Members







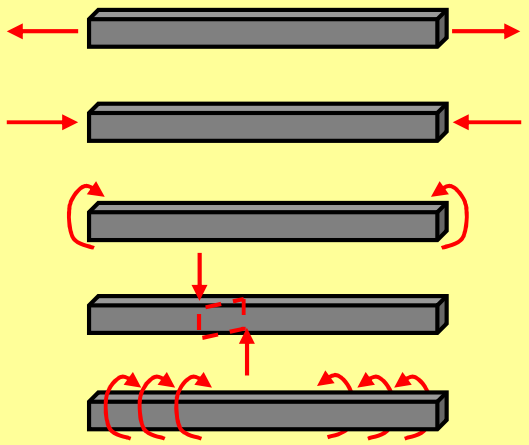
钢筋混凝土





钢筋混凝土基本构件

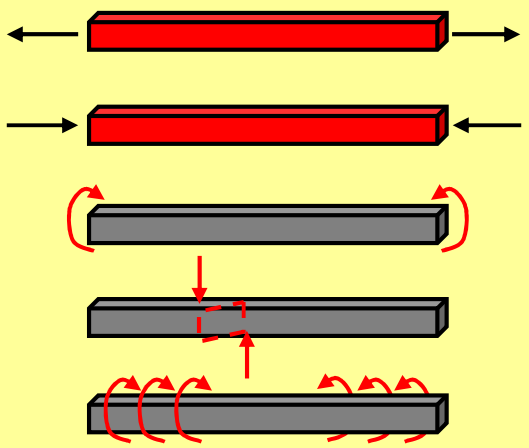
- 单一受力
  - 轴心受拉
  - 轴心受压
  - 正截面受弯
  - 斜截面受剪
  - 受扭



- 复合受力
  - 压-弯
  - 剪-扭
  - 弯-剪-扭

钢筋混凝土基本构件

- 单一受力
  - 轴心受拉
  - 轴心受压
  - 正截面受弯
  - 斜截面受剪
  - 受扭

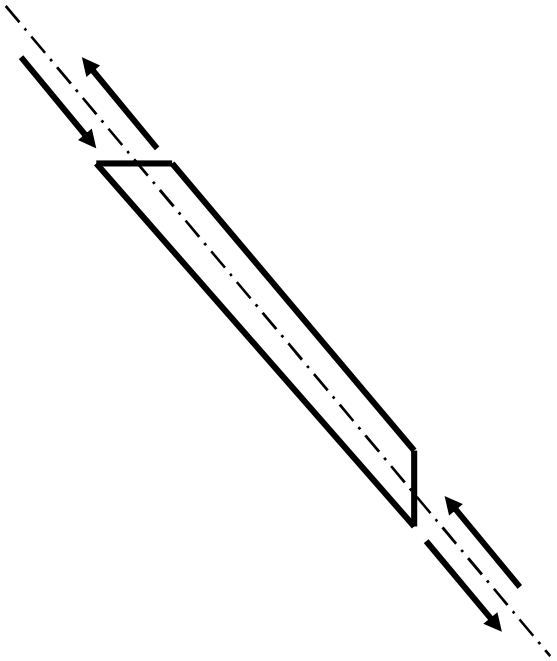


- 复合受力
  - 压-弯
  - 剪-扭
  - 弯-剪-扭

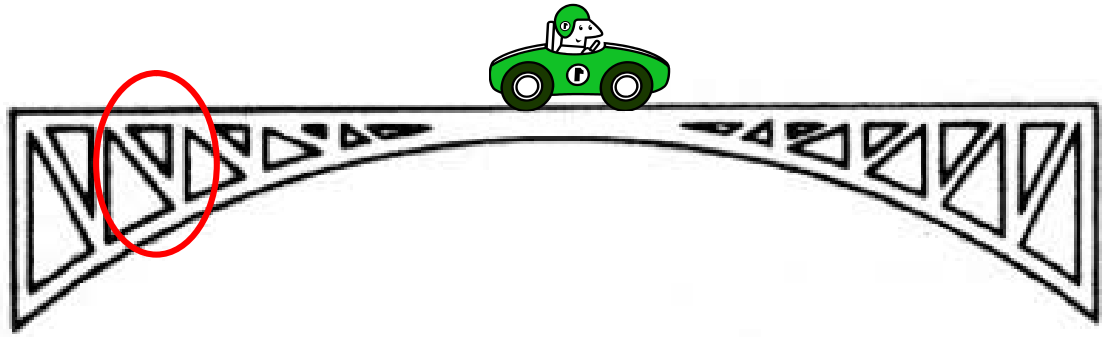
桥梁结构中的轴心受力构件

桁架杆件

Member in Truss



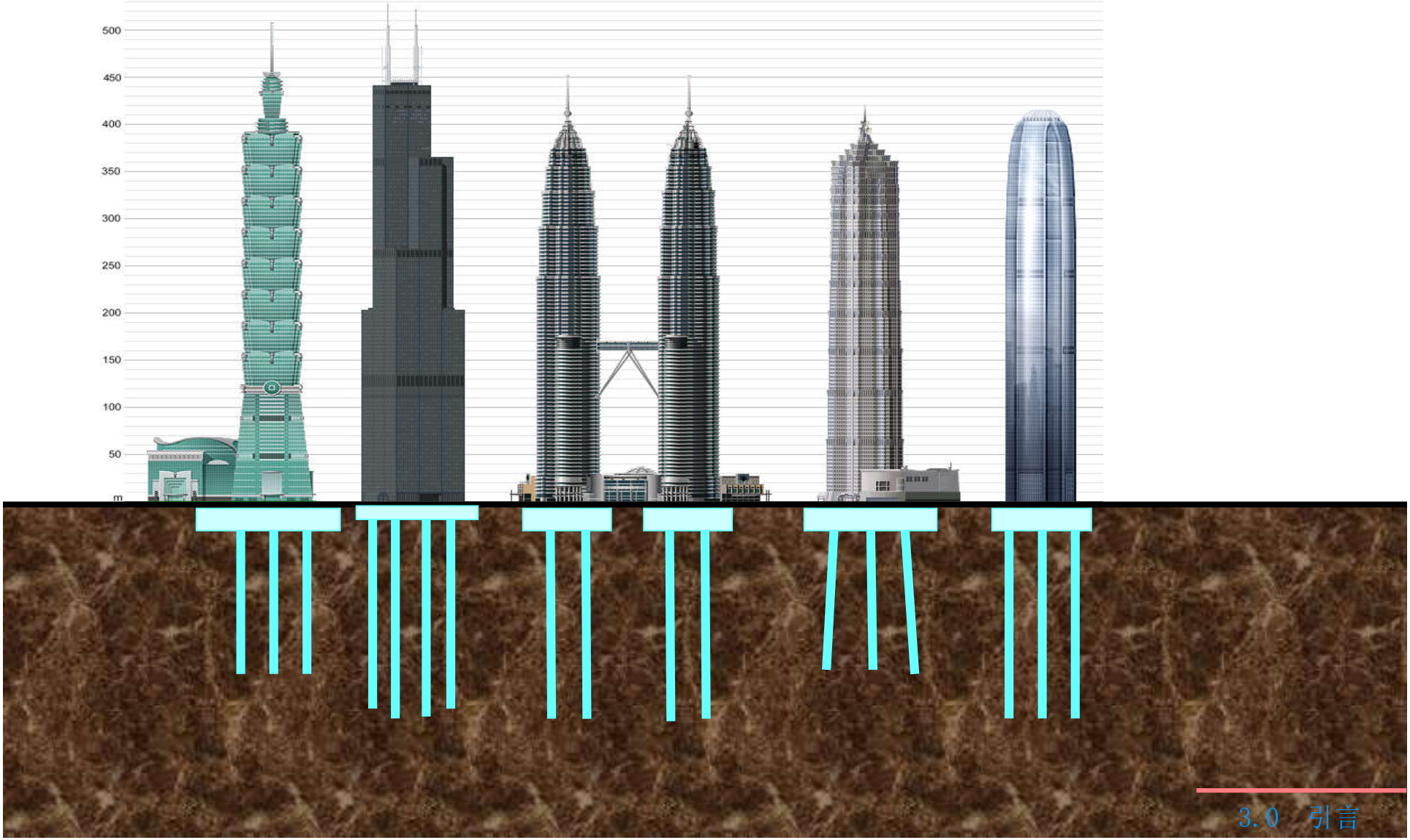




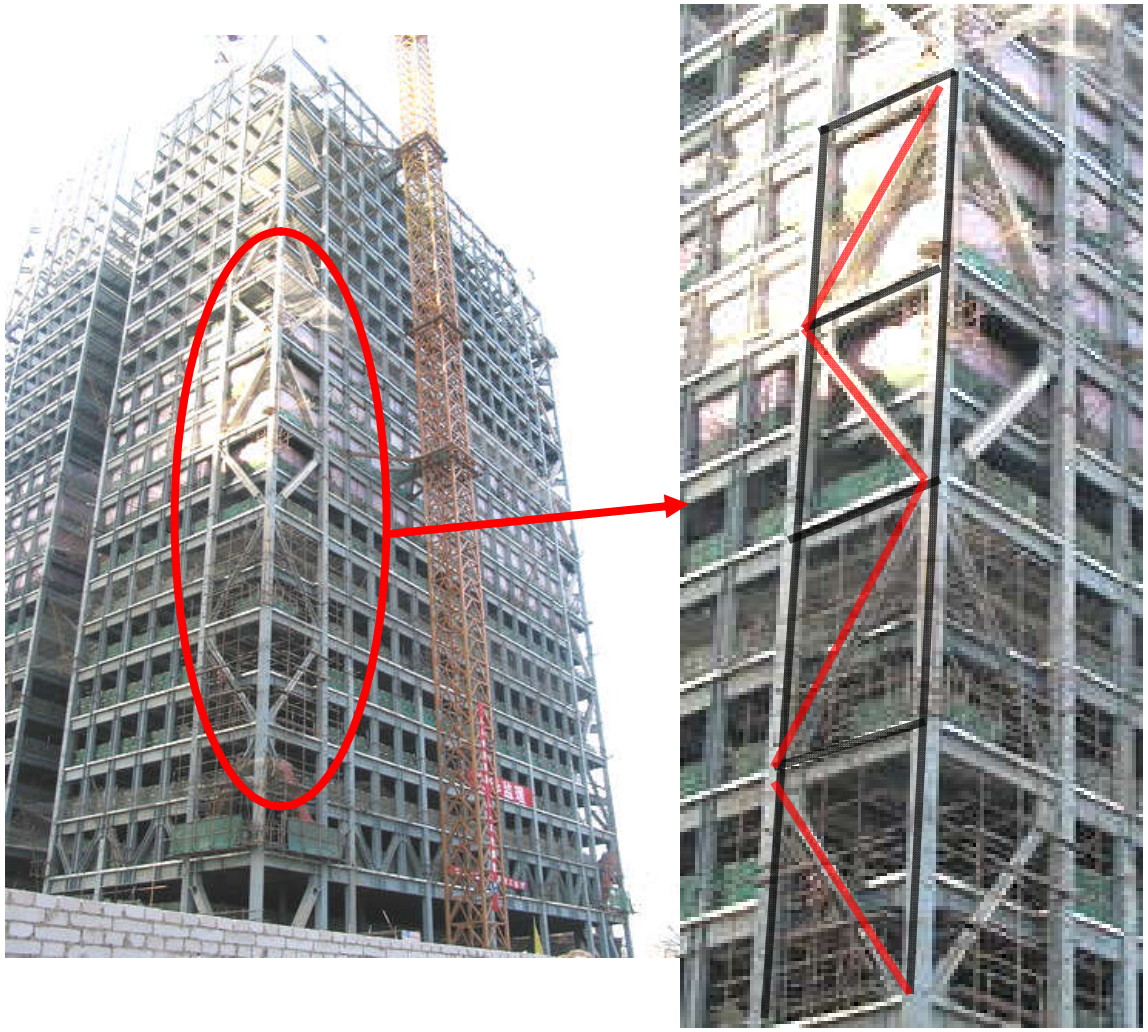
地下结构中的轴心受力构件

桩基础

Pile Foundation

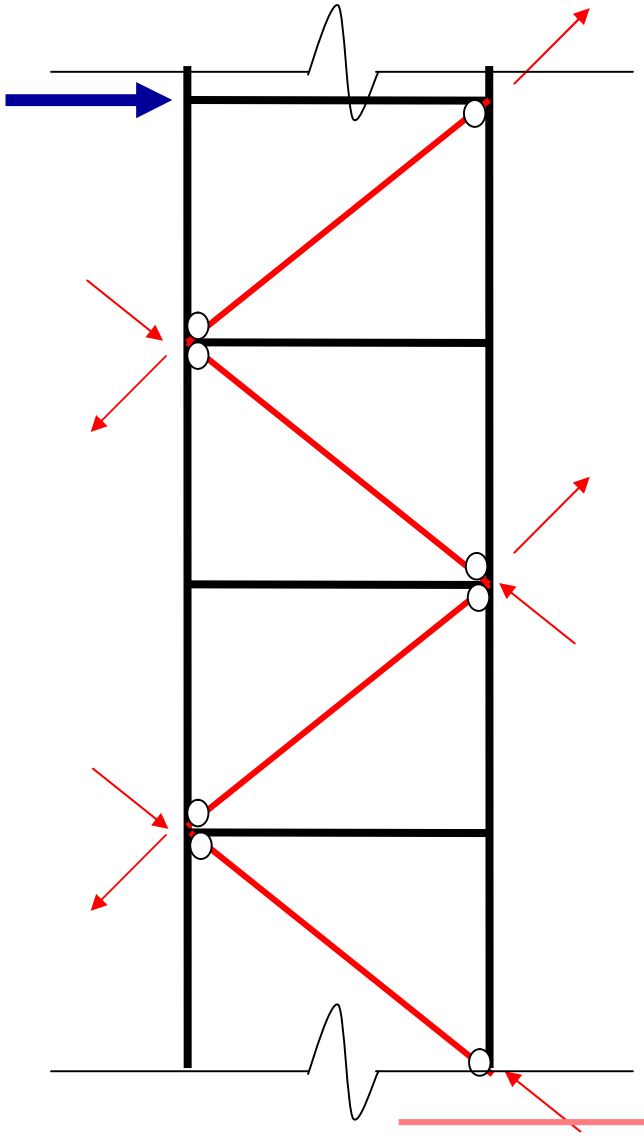


建筑结构中的轴心受力构件



斜向支撑

Diagonal Brace



最近研究　　WTC的倒塌


9:07AM Sep 11, 2001


9:30AM Sep 11, 2001

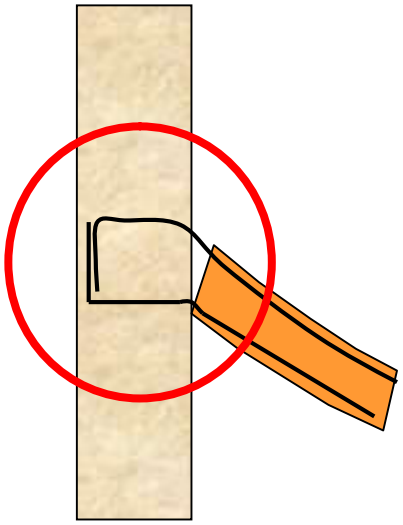

9:59AM Sep 11, 2001

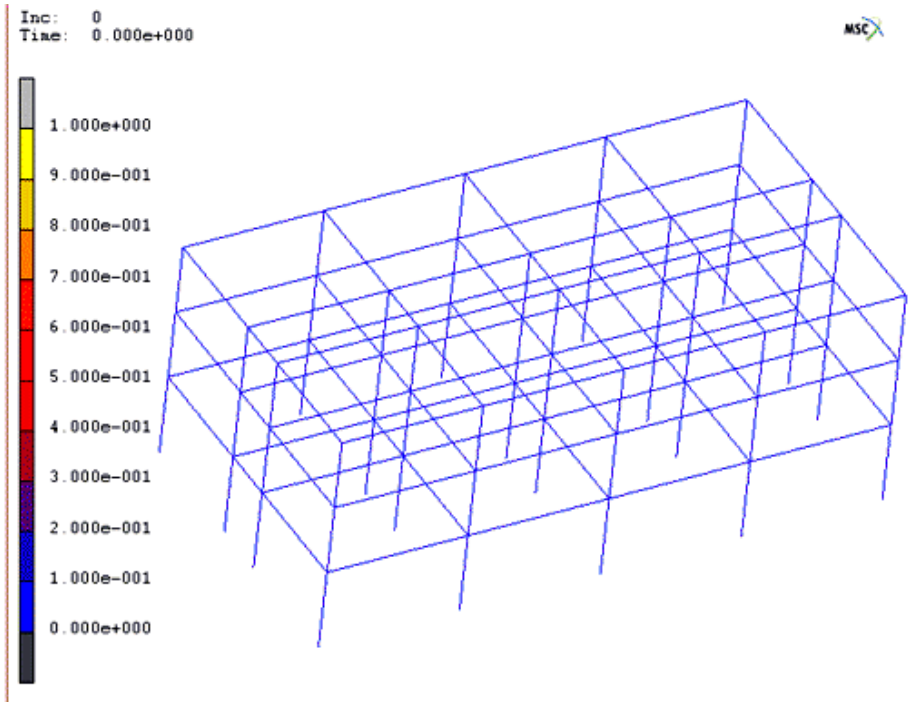

10:29AM Sep 11, 2001

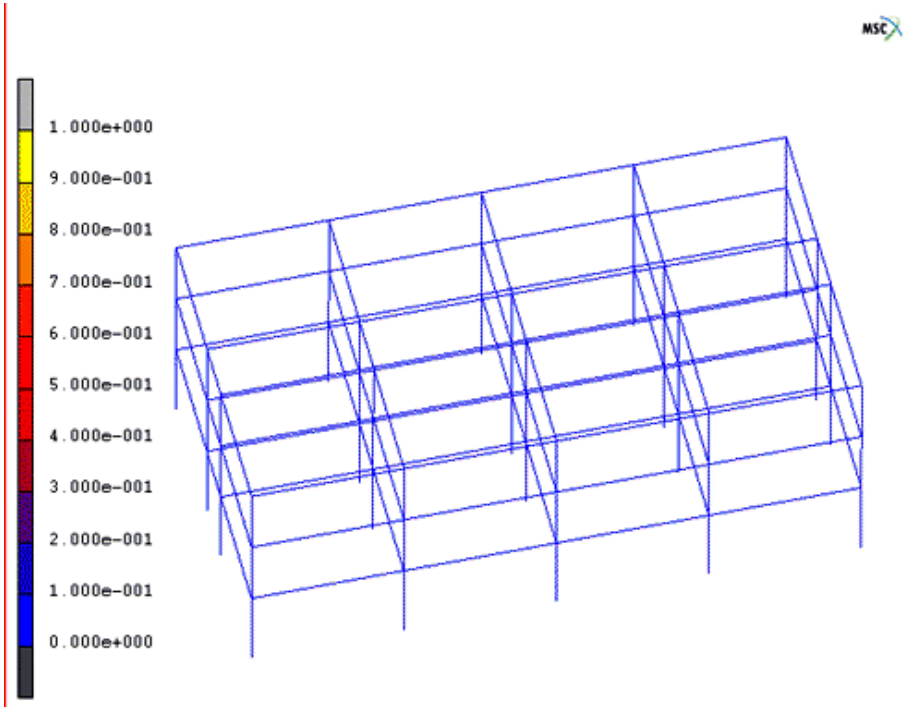
最近研究

防止连续倒塌的构件拉接

Anti-Collapsed Tie



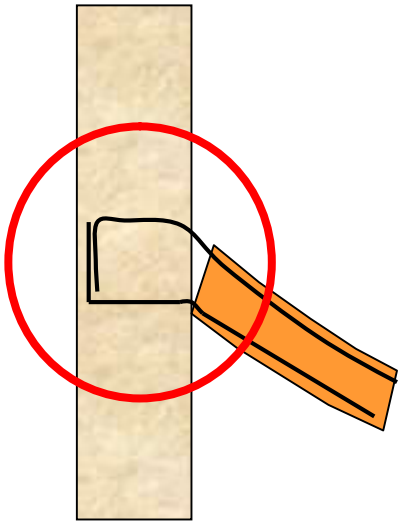




最近研究

防止连续倒塌的构件拉接

Anti-Collapsed Tie



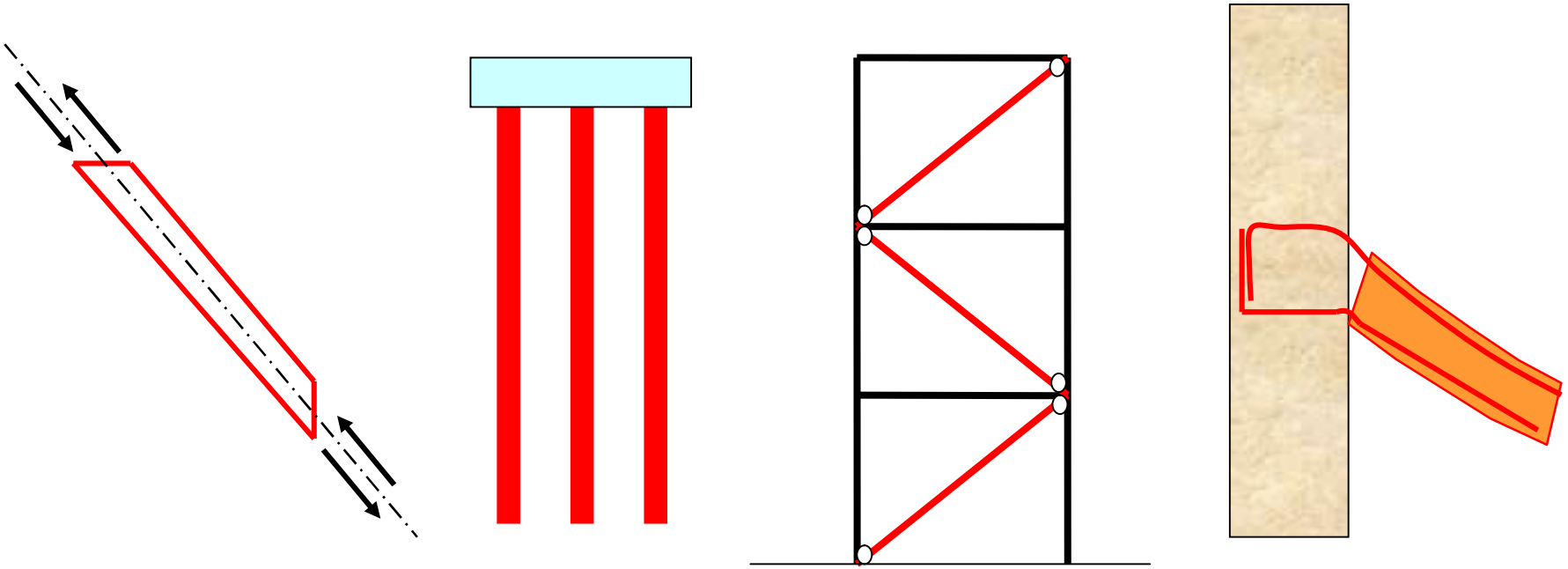








怎样用混凝土和钢筋组成满足受力要求的轴心受力构件？



3.0 引言

Introduction

3.1 轴心受拉构件的受力性能

Behavior of Uniaxial Tensile Member

3.2 轴心受拉构件的设计

Design on Uniaxial Tensile Member

3.3 轴心受压构件的受力性能

Behavior of Uniaxial Compressive Member

3.4 轴心受压构件的设计

Design on Uniaxial Compressive Member

3.0 引言

Introduction

3.1 轴心受拉构件的受力性能

Behavior of Uniaxial Tensile Member

3.2 轴心受拉构件的设计

Design on Uniaxial Tensile Member

3.3 轴心受压构件的受力性能

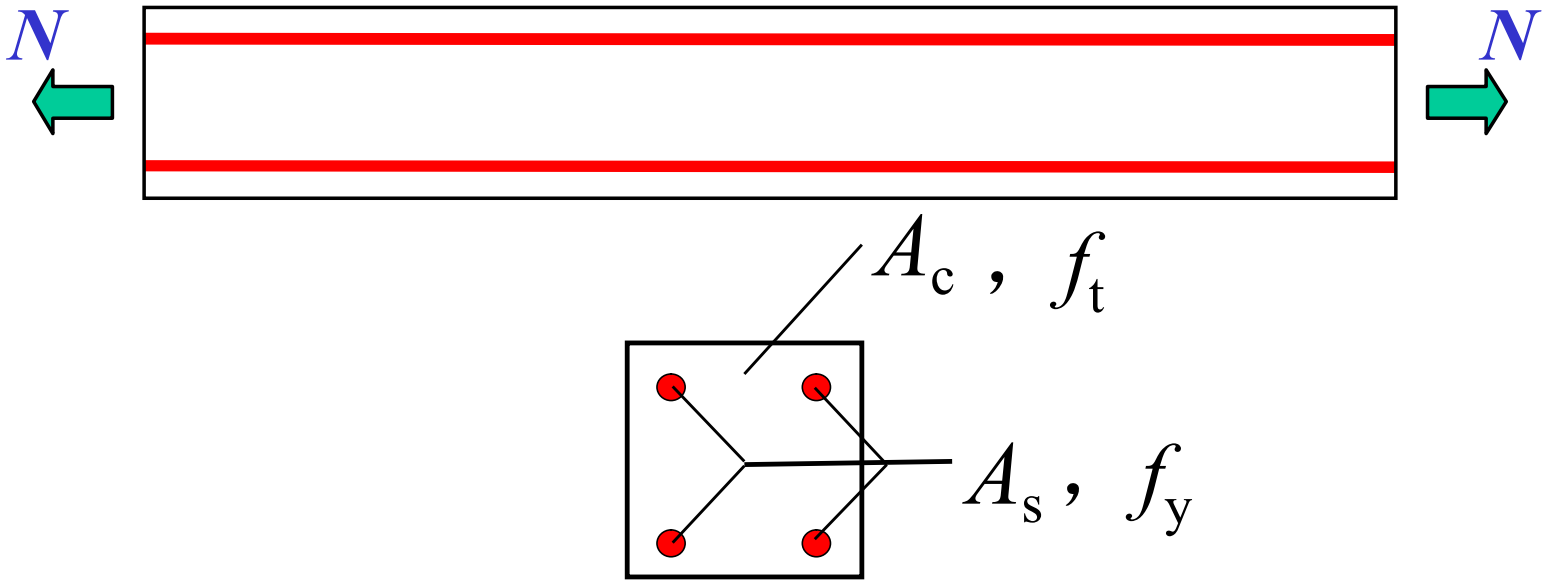
Behavior of Uniaxial Compressive Member

3.4 轴心受压构件的设计

Design on Uniaxial Compressive Member

3.1 轴心受拉构件的受力性能

Behavior of Uniaxial Tensile Member



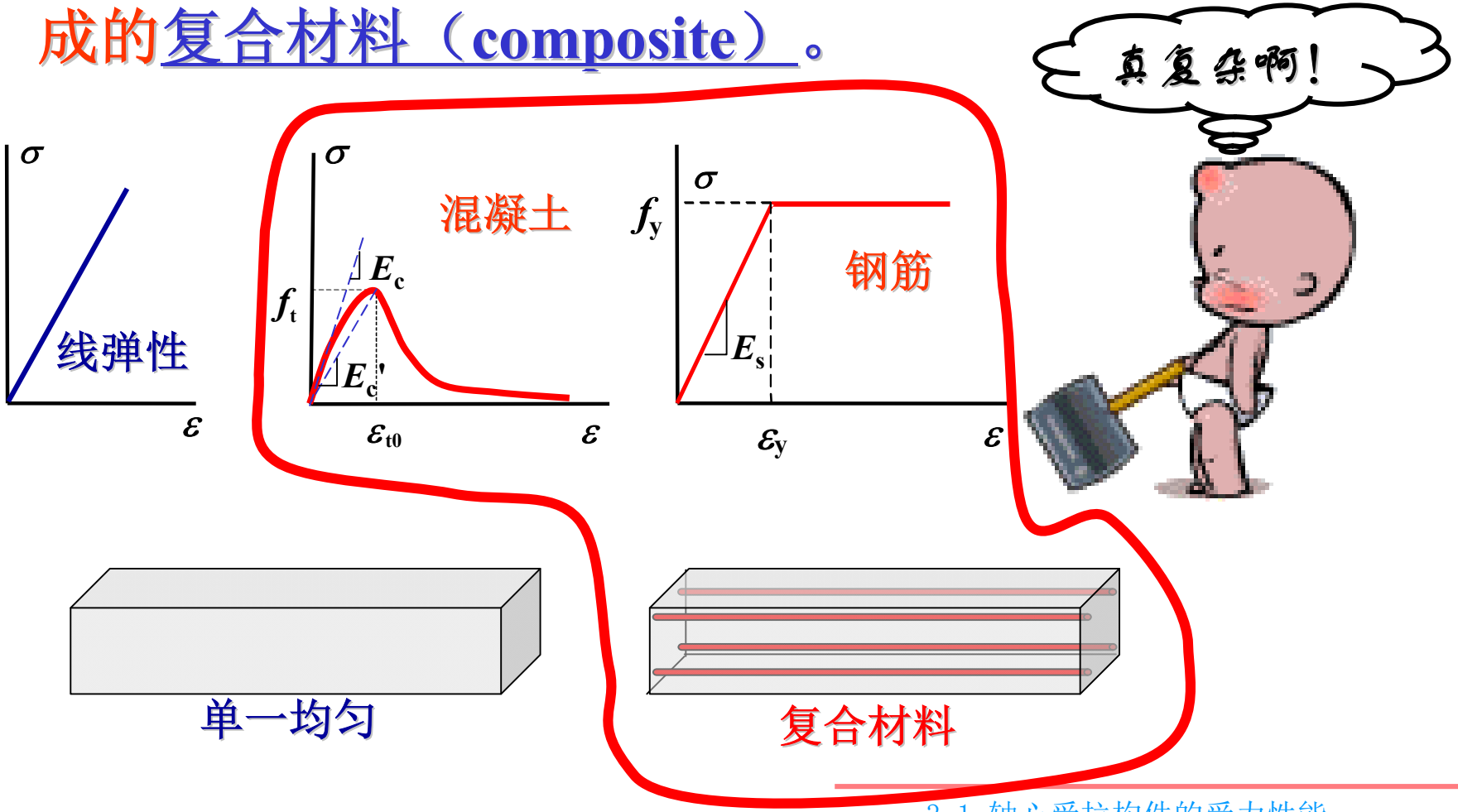
关键问题：

- 混凝土的强度和用量
- 钢筋的强度与用量

→ 受力性能

钢筋混凝土 Reinforced Concrete

两种非线性（nonlinear）材料组合而成的复合材料（composite）。



三句“秘诀”

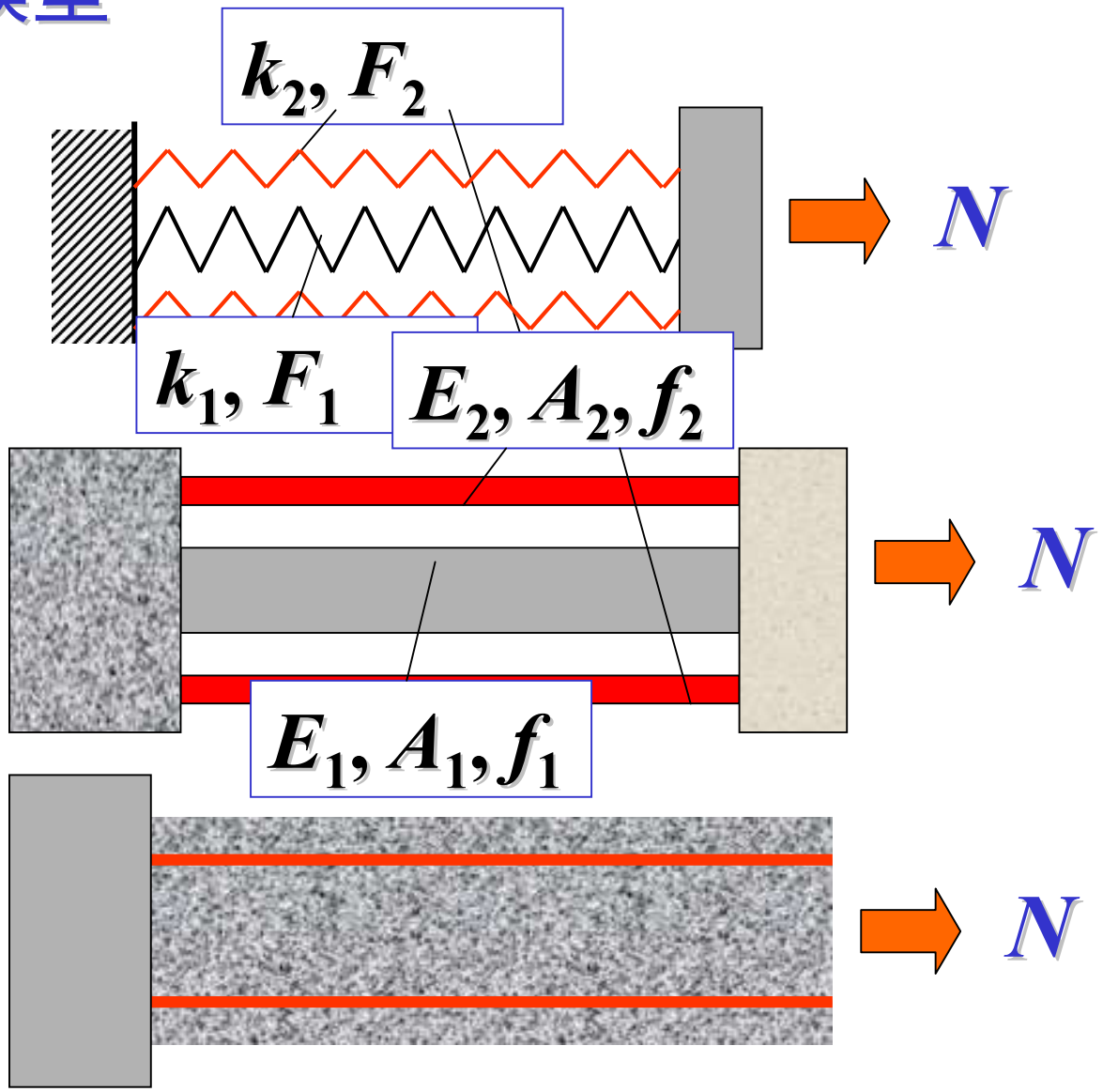
- 确定关键状态及其先后次序
- 确定前提条件：钢与混凝土共同工作 吗？
- 分阶段应用三大方程：
  平衡方程 equilibrium equations
  几何条件 consistent conditions
  物理关系 constitutive relations

轴心受力模型

几何条件

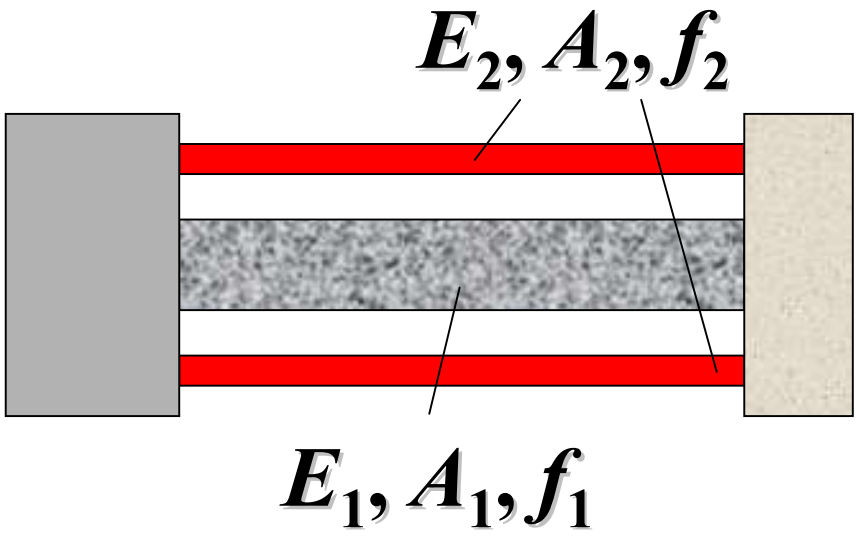
物理条件

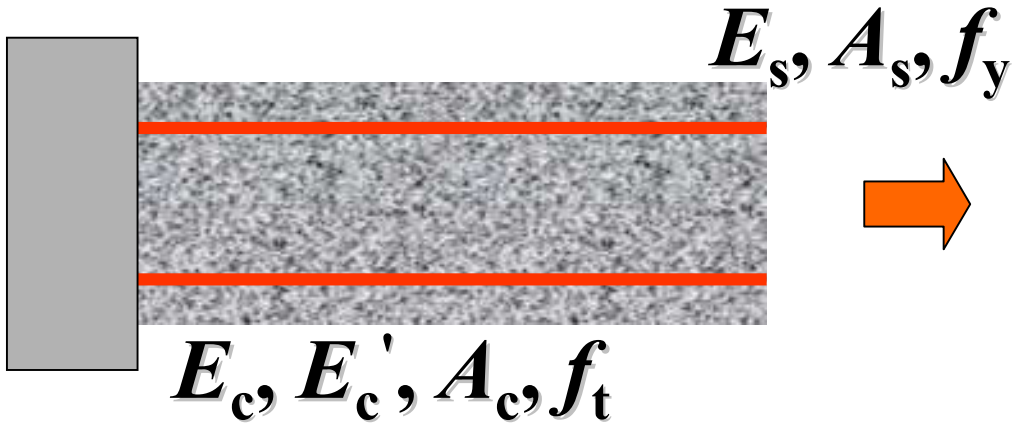
内力平衡

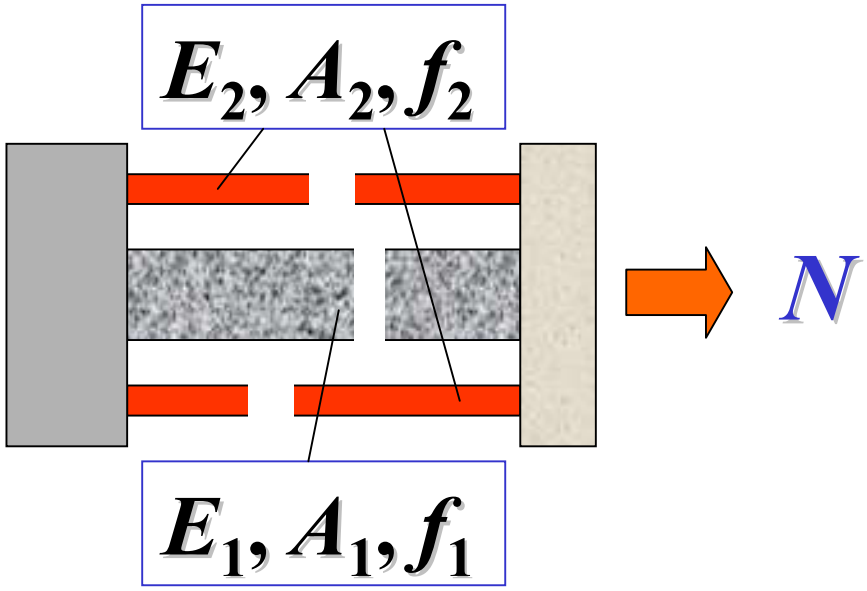


轴心受拉模型 Uniaxial Tensile Models

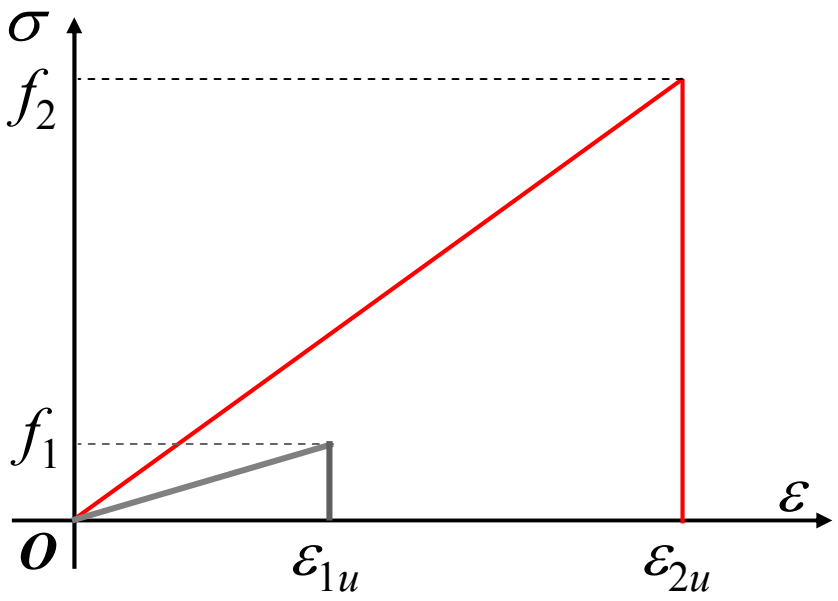
- 非线性
- 组合受力







- 两个关键状态：①、②
- 0 →①：两者共同工作；
- ①→②：材料2单独受力。



纤维复合材料筋　FRP筋


碳纤维


芳纶纤维


玻璃纤维

分析过程：

1. 确定关键状态
2. 分析共同工作条件
3. 应用三大方程



线弹性 ——————→ 非线性

请发送短信到 18618180598

题目：下列关于多种材料共同轴心受拉正确的有：
A 强度低的材料先发生破坏
B 弹性模量高的材料先发生破坏
C 极限承载能力可能只与其中某一材料的性能有关
D 极限变形能力一定只与其中某一材料的性能有关

我爱砼第4题“答案”2013010xxx

3.1 轴心受拉构件的受力性能

Behavior of Uniaxial Tensile Member

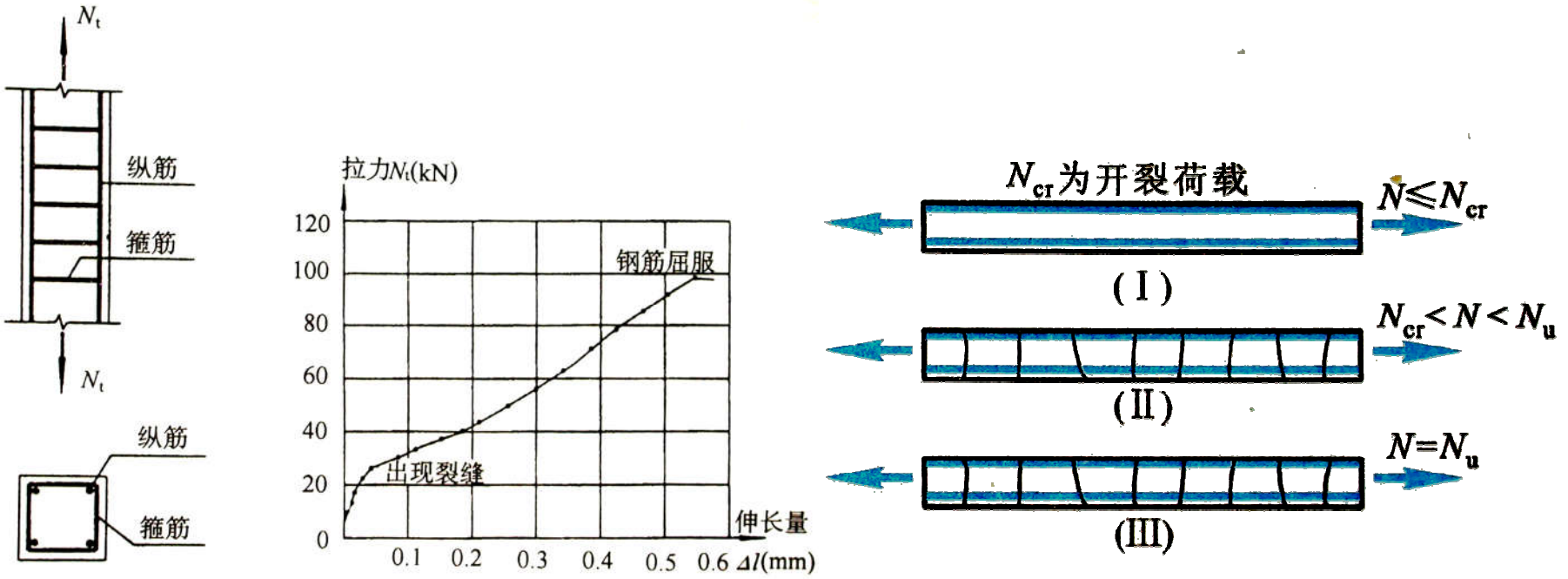
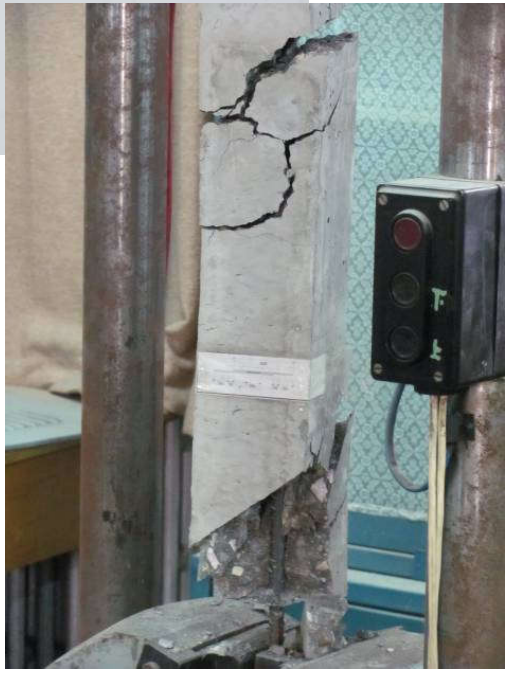


图 3－2　轴心受拉构件破坏的三个阶段



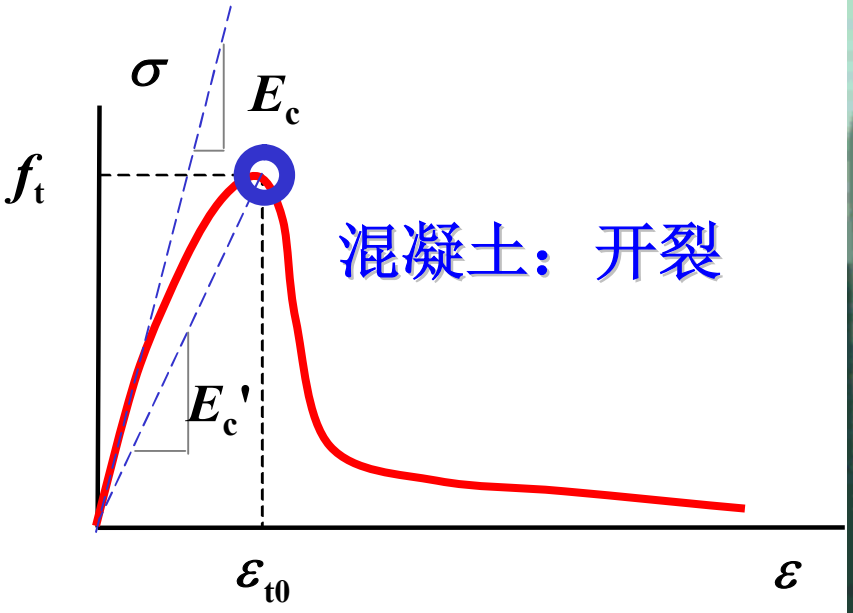


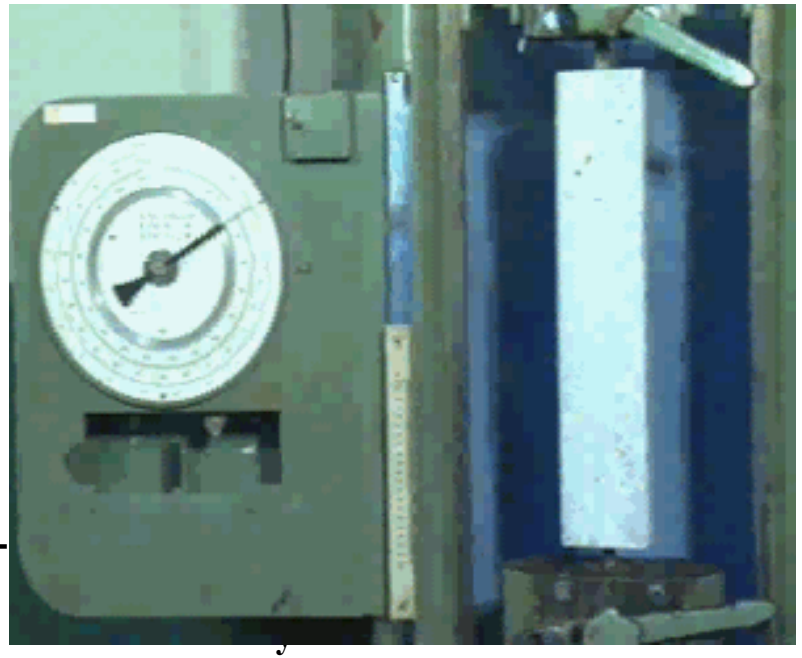




水平裂缝→端部斜裂缝→端部脱落→钢筋屈服→钢筋颈缩→破坏

受拉应力-应变关系 Tensile Stress-Strain Relationships





$$\varepsilon_{t0}=\frac{f_t}{E_c'}=\frac{f_t}{0.5E_c}=\frac{2f_t}{E_c}$$

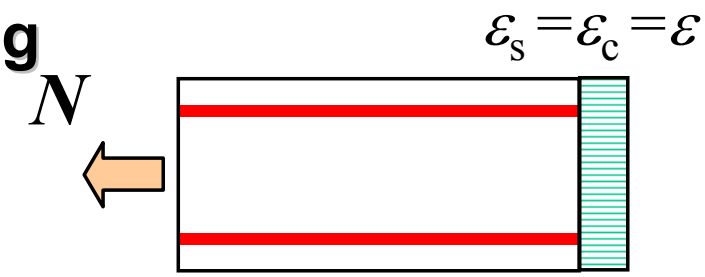
割线模量

$$\sigma=E_s\varepsilon \qquad \varepsilon\le\varepsilon_y$$
$$\sigma=f_y \qquad \varepsilon>\varepsilon_y$$

$$\varepsilon_{t0}<<\varepsilon_y$$

150～300$\mu\varepsilon$　　1200～2000$\mu\varepsilon$

◆开裂前 $N<N_{cr}$ Before Cracking



◎几何条件consistent condition

$$\varepsilon_s=\varepsilon_c=\varepsilon$$

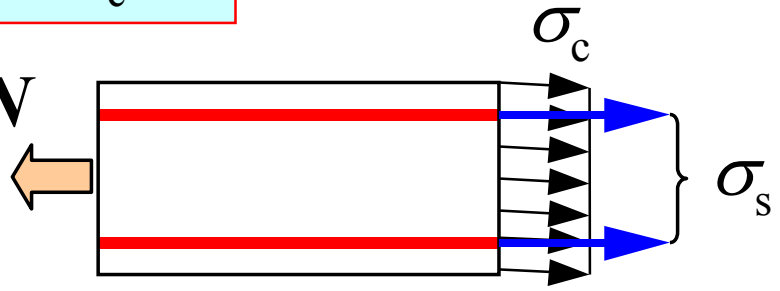
◎物理关系 stress-strain relationship

$$\sigma_s=E_s\varepsilon_s=E_s\varepsilon$$

$$\sigma_c=E'_c\varepsilon_c=\nu E_c\varepsilon$$

$$\Rightarrow \quad \sigma_s=\frac{E_s}{\nu E_c}\sigma_c$$

◎平衡方程 equilibrium equation



$$N=\sigma_c A_c+\sigma_s A_s$$

$$=\sigma_c A_c\left(1+\frac{E_s}{E'_c}\frac{A_s}{A_c}\right)=\sigma_c A_c\left(1+\frac{E_s}{\nu E_c}\frac{A_s}{A_c}\right)=\sigma_c A_c\left(1+\rho\frac{\alpha_E}{\nu}\right)$$

$$\rho=A_s/A_c \qquad \alpha_E=E_s/E_c$$

◆ 开裂荷载 Cracking Load

当$\sigma_c = f_t$时，如继续加载混凝土将会产生开裂，达到N_{cr}

由混凝土的受拉时应力-应变关系知，$\sigma_c = f_t$ 时的弹性特征系数ν=0.5

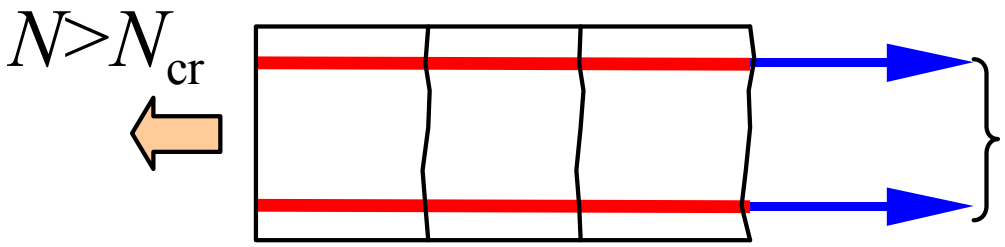
$$\sigma_s = \frac{\alpha_E}{\nu}\sigma_c = \frac{\alpha_E}{0.5} f_t = 2\alpha_E f_t$$

$$N_{cr} = f_t A_c + \sigma_s A_s = f_t A_c \left(1 + 2\alpha_E \rho\right)$$

◆混凝土开裂后 After Cracking

如果配筋足够，荷载可以继续增加

但裂缝截面混凝土应力始终为零，开裂以后的全部荷载增量均由钢筋承担



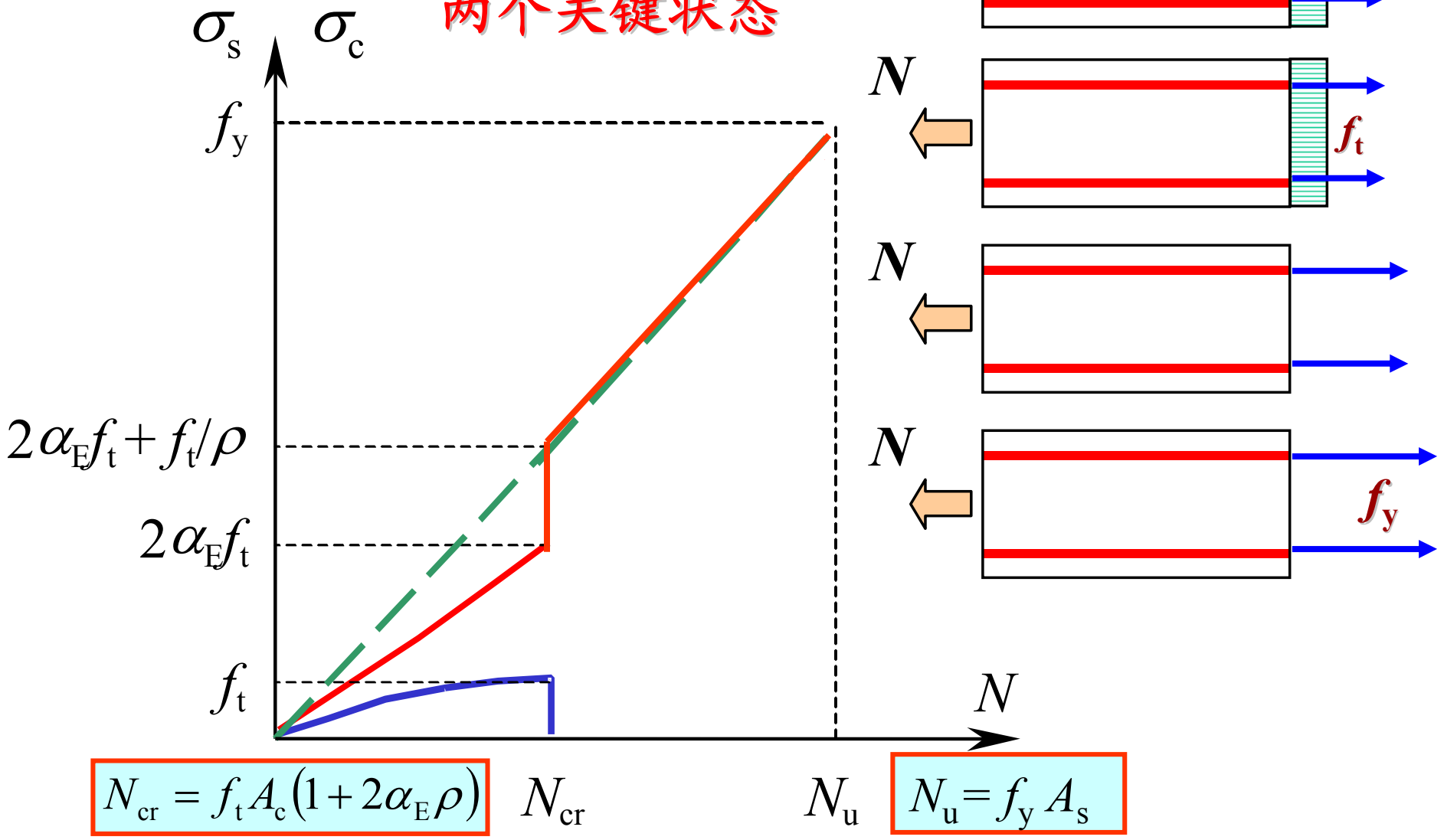
当钢筋应力达到$\sigma_s = f_y$时

极限拉力Ultimate Tensile Capacity

$$N_u = f_y A_s$$

小 结 Summary

两个关键状态



$N_{cr} = f_t A_c (1 + 2\alpha_E \rho)$

$N_u = f_y A_s$

小 结 Summary

- 三句“秘诀”
- 两个处理非线性问题的方法
- 钢筋混凝土构件分析
  vs. 材料力学分析
  复杂对象
  vs. 简单方法

讨论

(1) 由于混凝土材料的非线性和开裂，荷载与应力、荷载与变形不再为线性关系，线弹性材力分析方法不再适用；

(2) 混凝土开裂前瞬间，钢筋应力=20~40MPa，对N_{cr}影响很小。

(3) 在混凝土开裂后瞬间，截面应力发生重分布；

(4) 开裂后，构件是带裂缝工作的。

裂缝对钢筋受力的影响？

裂缝宽度的计算？

裂缝对变形的影响（刚度）？

小 结 Conclusions

(5) RC构件中钢筋的应变小于同样面积的钢筋单独受拉情况（开裂前钢筋与混凝土共同承受拉力，开裂后裂缝间混凝土仍承受一部分拉力）；

(6) RC轴心受拉构件的受力过程大体上可分三个阶段

①开裂前的钢筋与混凝土的共同受力阶段，该阶段荷载-变形关系基本近似线性；

②开裂后带裂缝工作阶段；

③破坏阶段。

★ 对于配筋合适的钢筋混凝土构件，一般其受力过程都可分为类似的三个工作阶段。

轴心受拉构件的设计

$$N \leqslant f_{\mathrm{y}} A_{\mathrm{s}}$$

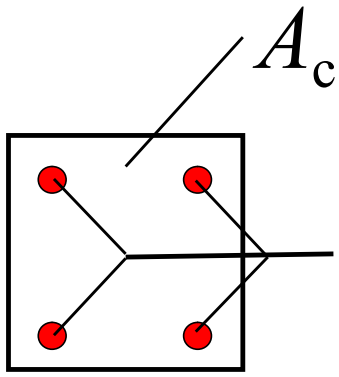
构造要求：

（1）纵向受力钢筋不得采用绑扎搭接接头

（2）最小配筋率

（3）受力钢筋沿周边均匀布置，优先直径小的钢筋

配筋率的影响



$$\rho = A_s / A_c$$

开裂前？　$\sigma_s = 2\alpha_E f_t = 20 \sim 40 MPa$

开裂后？　$\sigma_s = \dfrac{N_{cr}}{A_s} = 2\alpha_E f_t + \dfrac{f_t}{\rho}$

应力重分布

如果配筋率ρ小于某一界限时，以至于在混凝土一开裂时，由于混凝土退出工作产生应力重分布，使钢筋立刻就达到屈服，即，

$$N_u = N_{cr}$$

当配筋率小于该界限配筋率时，钢筋将不能承担混凝土截面开裂后传来的应力，其破坏特征如同素混凝土构件，具有脆性性质。因此RC轴拉构件的配筋率应不小于该界限配筋率

◆**最小配筋率** Minimum Reinforcement Ratio

$$f_y A_s = f_t A_c \left(1 + 2\alpha_E \rho\right)$$

$$\rho_{\min} = \frac{f_t}{f_y - 2\alpha_E f_t}$$

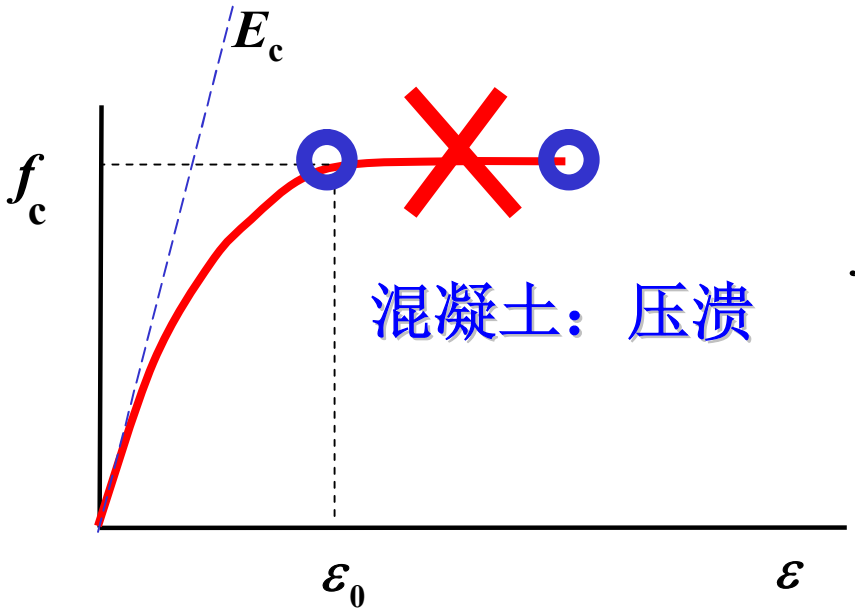
3.3 轴心受压构件的受力性能

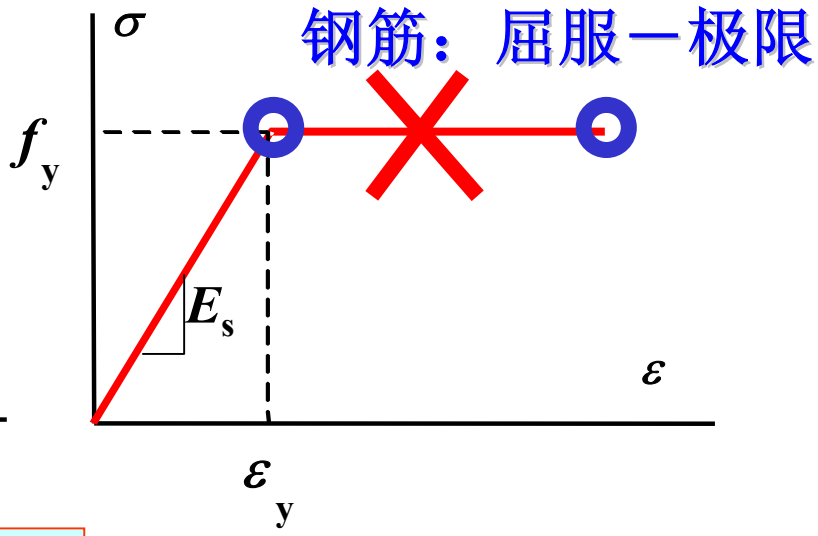
Behavior of Uniaxial Compressive Member

RC构件轴心受压的分析方法，与轴心受拉有什么不同？

从工程应用考虑，需要注意什么问题？

受压应力-应变关系



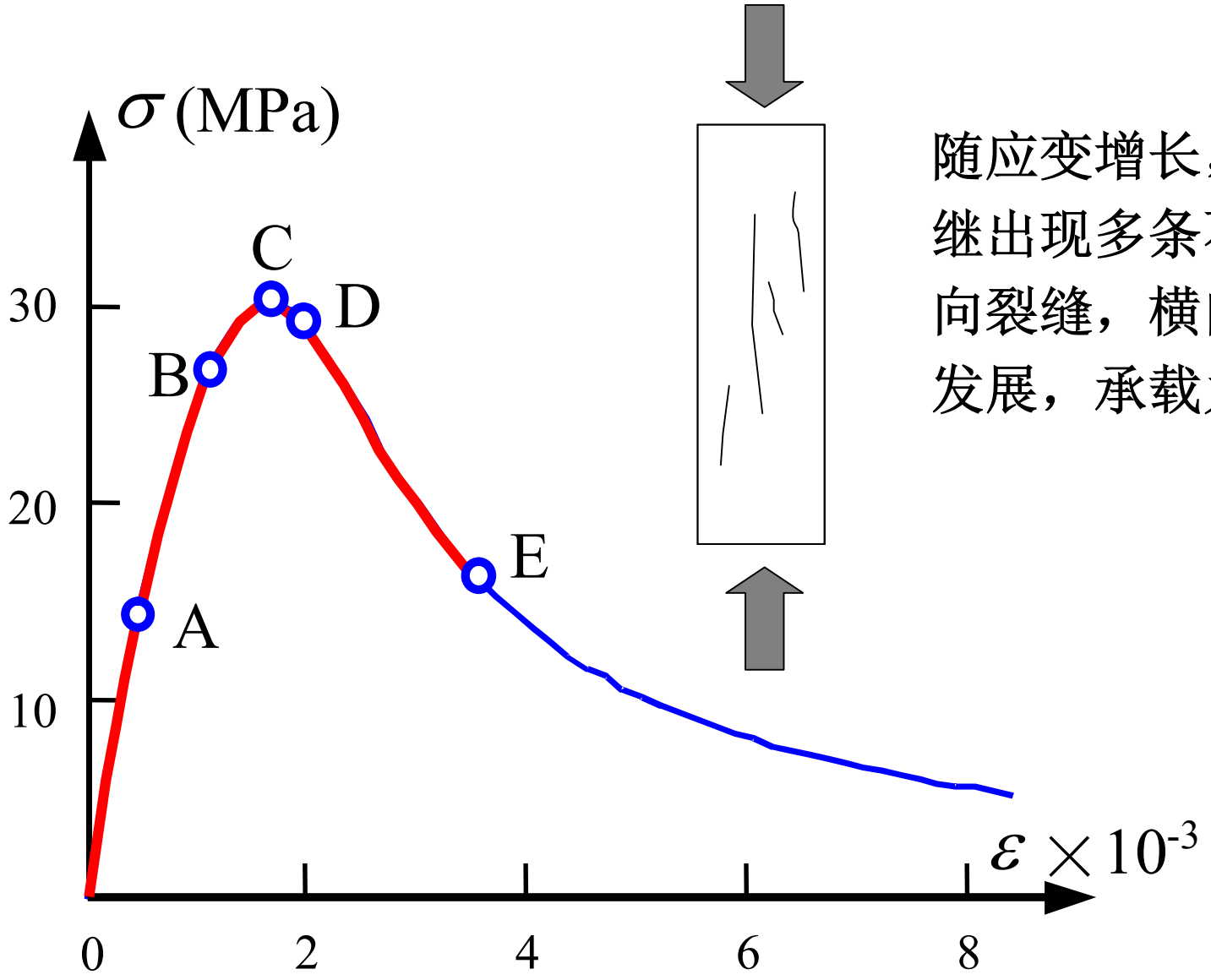


上升段：$\sigma_c = f_c[1-(1-\frac{\varepsilon_c}{\varepsilon_0})^n] \quad \varepsilon \le \varepsilon_0$

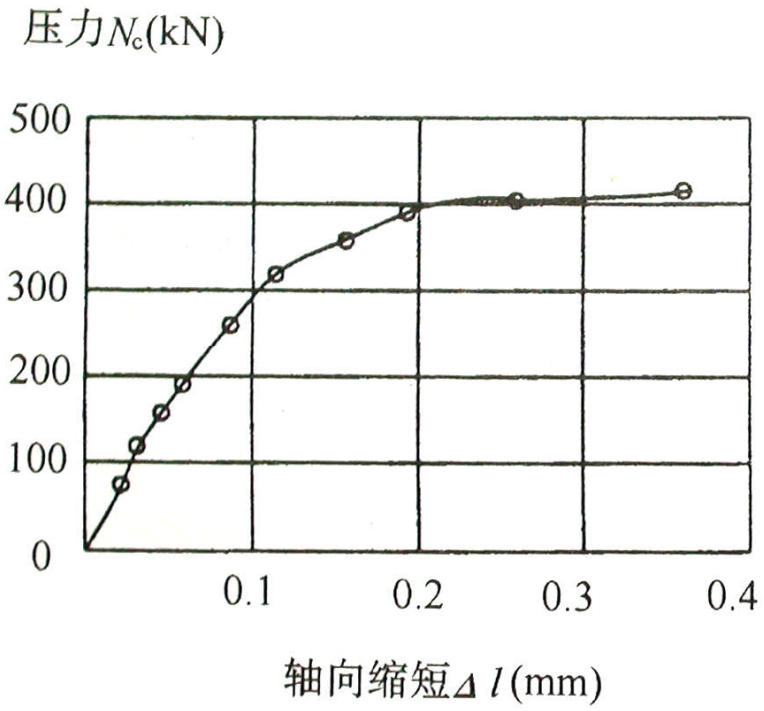
下降段：$\sigma_c = f_c \qquad \varepsilon_0 < \varepsilon \le \varepsilon_u$

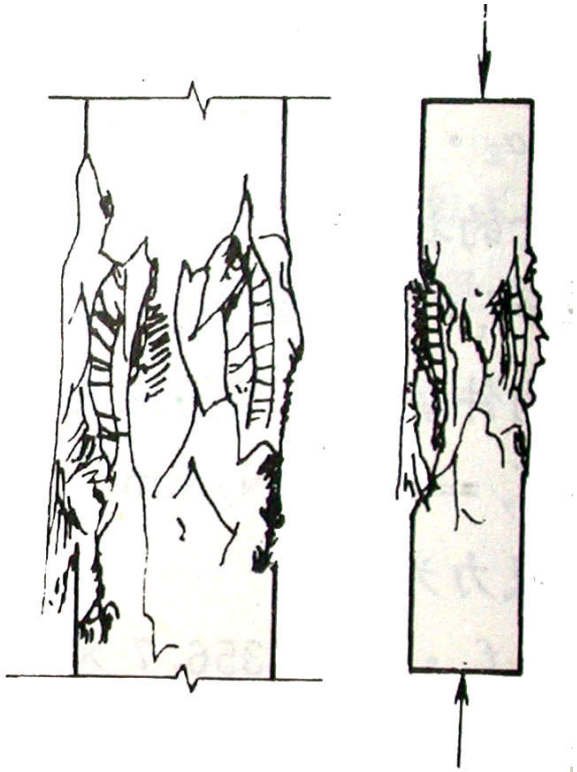
$\sigma = E_s\varepsilon \qquad \varepsilon \le \varepsilon_y$

$\sigma = f_y \qquad \varepsilon > \varepsilon_y$



随应变增长，试件上相继出现多条不连续的纵向裂缝，横向变形急剧发展，承载力明显下降


压力N_c(kN)
500
400
300
200
100
0
0.1
0.2
0.3
0.4
轴向缩短Δl(mm)



3.2 轴心受压构件的受力性能

Behavior of Uniaxial Compressive Member

RC构件轴心受压的分析方法，与轴心受拉有什么不同？

从工程应用考虑，需要注意什么问题？

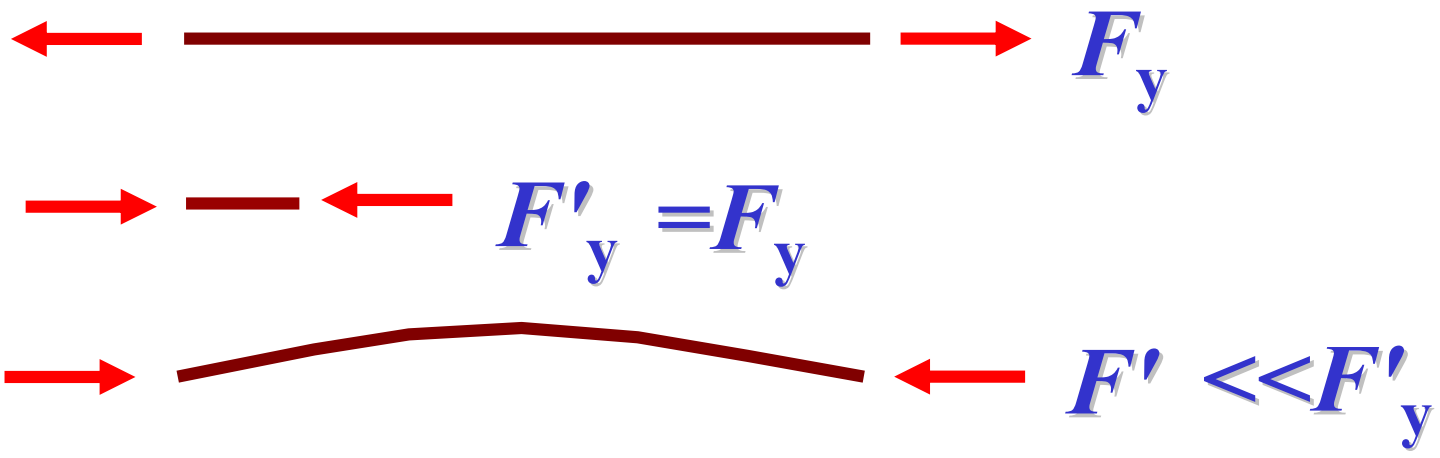
◆ 钢 材

◎ 抗拉和抗压强度都很高

Both tensile and compressive strengths are high

◎ 具有屈服现象，破坏时表现出较好的延性 Ductility

◎ 但细长的钢筋受压时极易压曲





刘立新提供



来源：不详





照片：台湾集集大地震　蔡万来 提供

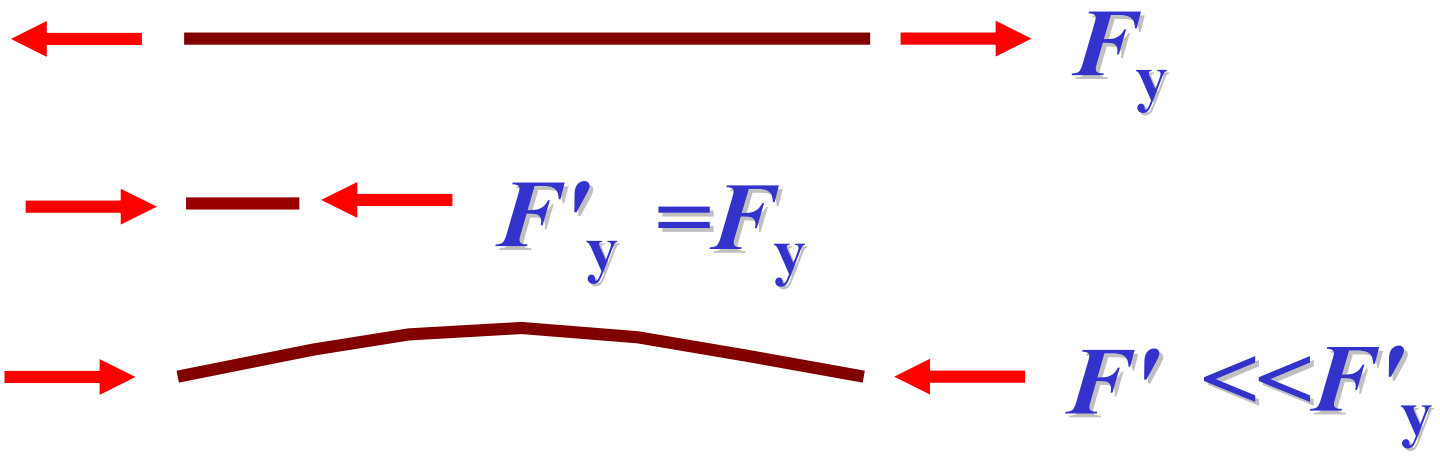
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◎ 但细长的钢筋受压时极易压曲



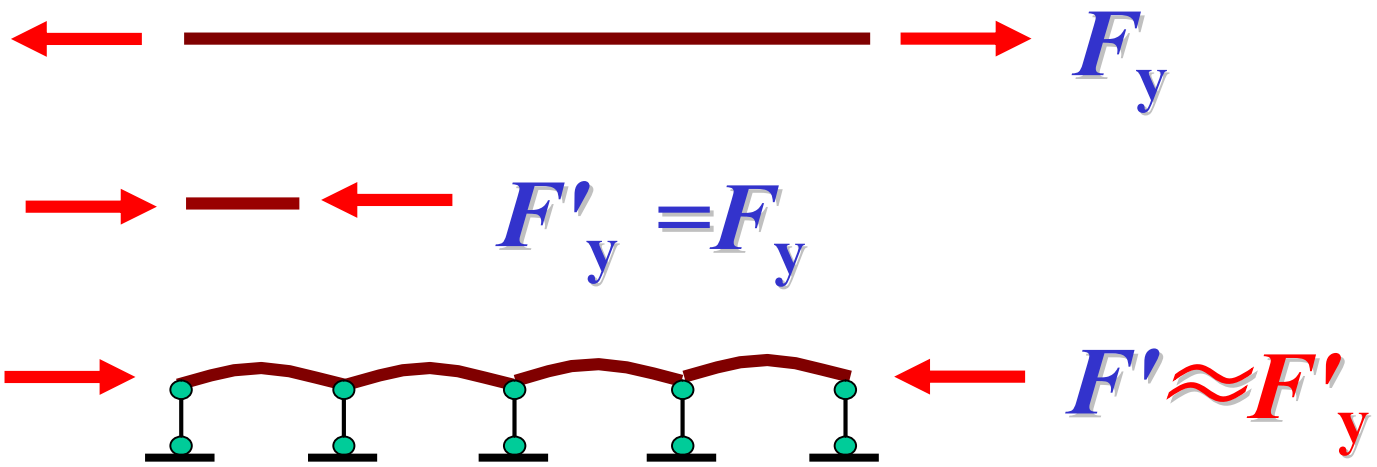
◆ 钢 材

◎ 抗拉和抗压强度都很高

Both tensile and compressive strengths are high

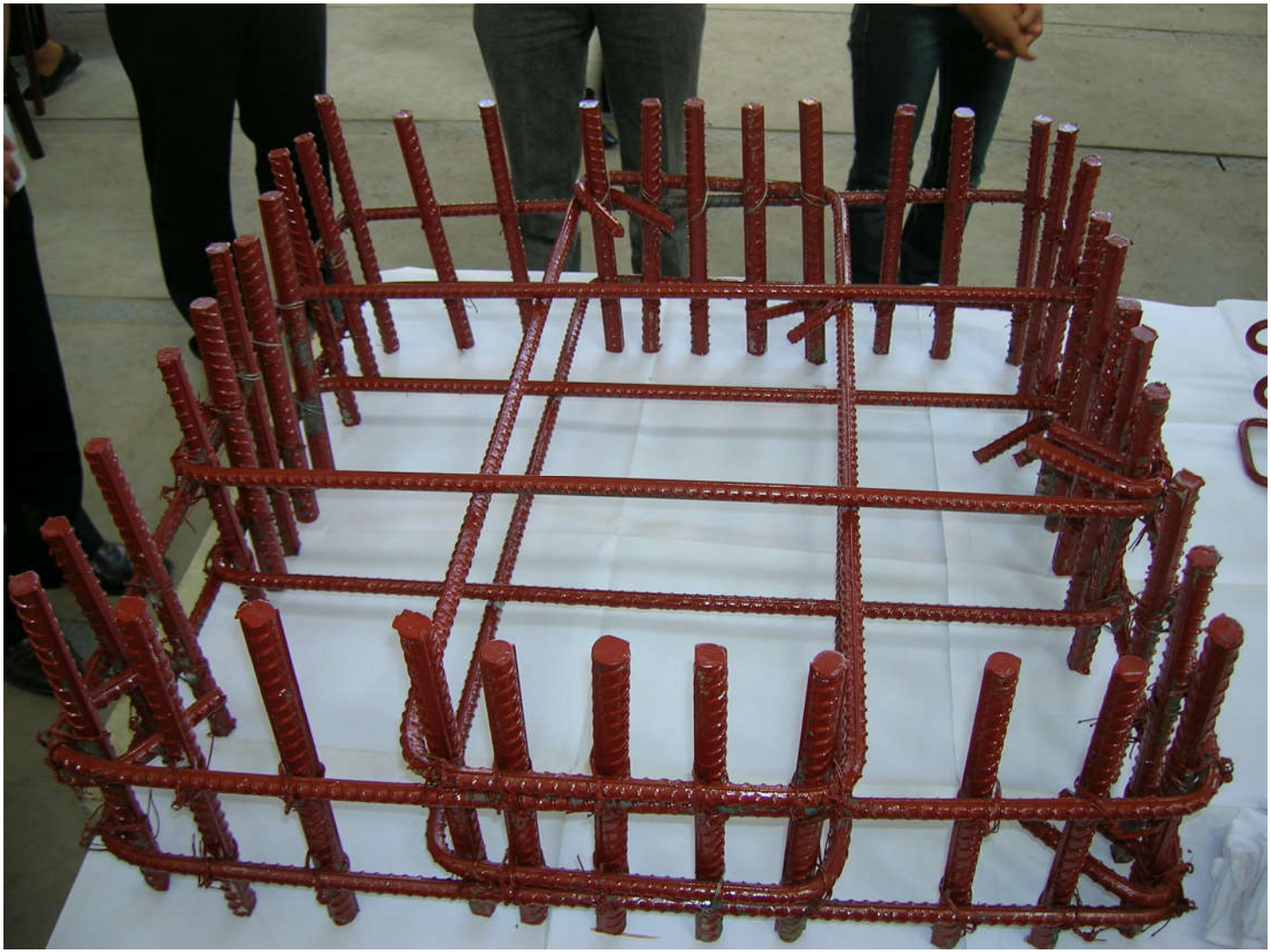
◎ 具有屈服现象，破坏时表现出较好的延性 Ductility

◎ 但细长的钢筋受压时极易压曲










A

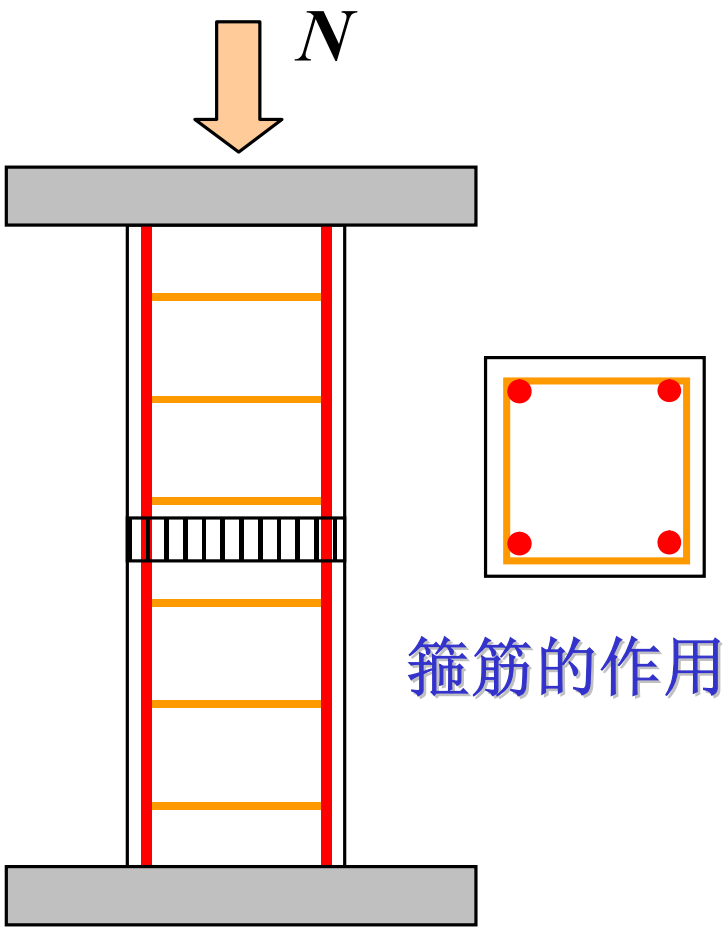






2 轴心受压构件的受力性能Behavior of Axial Compressive Member

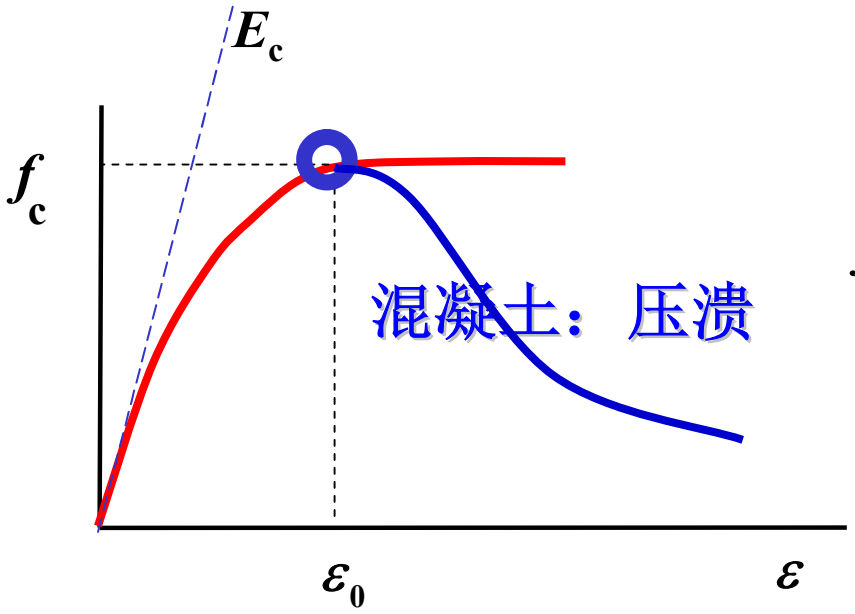
矩形截面轴心受压短柱

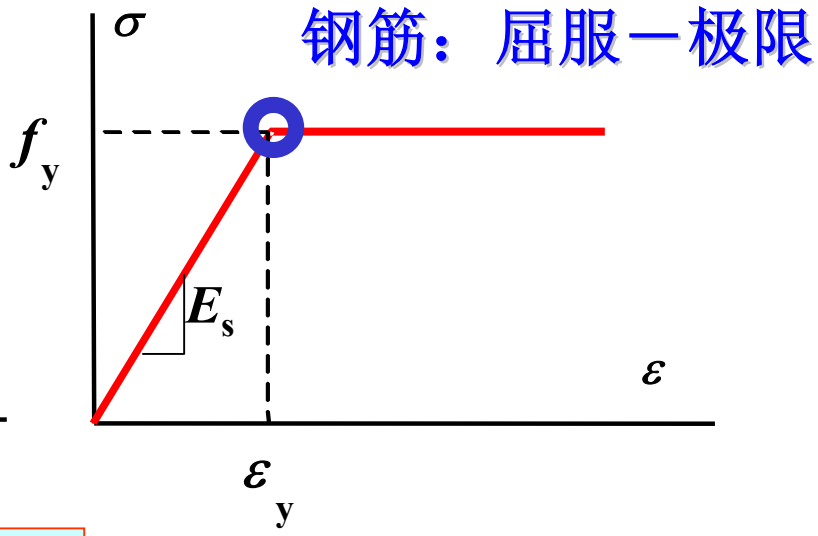


箍筋的作用？

Transverse reinforcement, ties or hoops, shall be provided throughout the height of a column to provide confinement and to prevent buckling of the longitudinal bars.

受压应力-应变关系





上升段：$\sigma_c = f_c[1-(1-\frac{\varepsilon_c}{\varepsilon_0})^n] \quad \varepsilon \le \varepsilon_0$

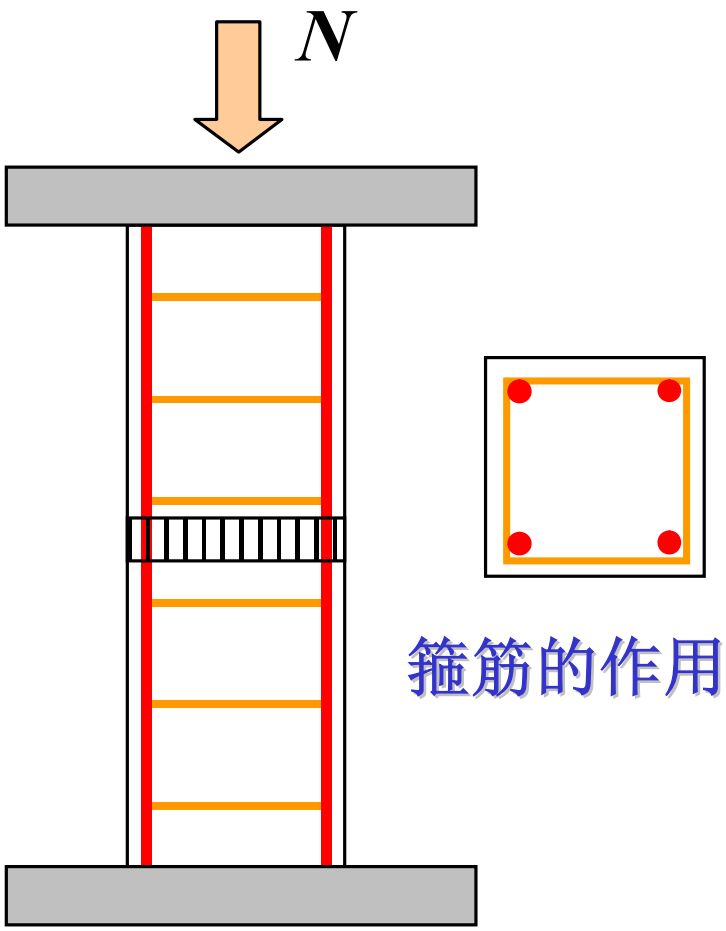
下降段：$\sigma_c = f_c \qquad \varepsilon_0 < \varepsilon \le \varepsilon_u$

$\sigma = E_s\varepsilon \qquad \varepsilon \le \varepsilon_y$

$\sigma = f_y \qquad \varepsilon > \varepsilon_y$

2 轴心受压构件的受力性能Behavior of Axial Compressive Member

矩形截面轴心受压短柱



箍筋的作用？

变形条件：$\varepsilon_s = \varepsilon_c = \varepsilon$

物理关系：$\sigma_s = E_s \varepsilon \qquad \varepsilon \le \varepsilon_y = \dfrac{f_y}{E_s}$

$\sigma_s = f_y \qquad \varepsilon > \varepsilon_y$

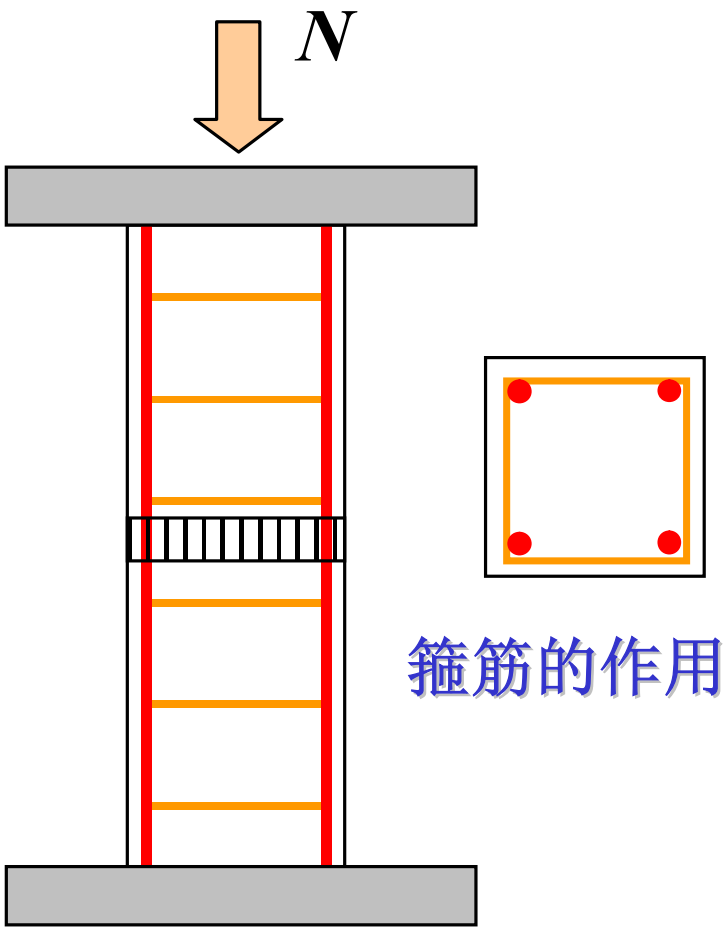
$$\sigma_c = f_c \left[\frac{2\varepsilon}{\varepsilon_0} - \left(\frac{\varepsilon}{\varepsilon_0} \right)^2 \right] \qquad 0 \le \varepsilon \le \varepsilon_0$$

平衡条件：

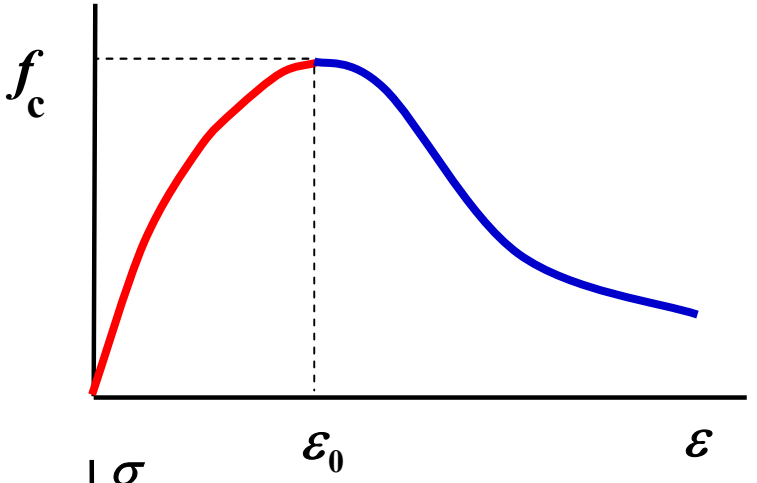
$$N = \sigma_c A_c + \sigma_s A_s$$

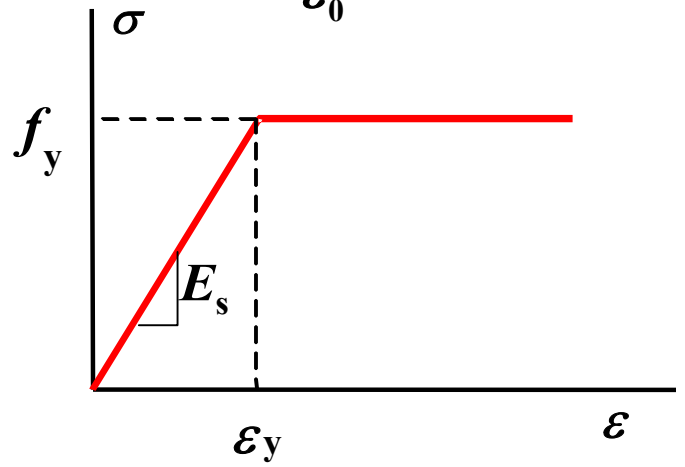
2 轴心受压构件的受力性能Behavior of Axial Compressive Member

矩形截面轴心受压短柱



箍筋的作用？





$$N = \sigma_c A_c + \sigma_s A_s$$

2 轴心受压构件的受力性能Behavior of Axial Compressive Member

矩形截面轴心受压短柱

极限轴压承载力？

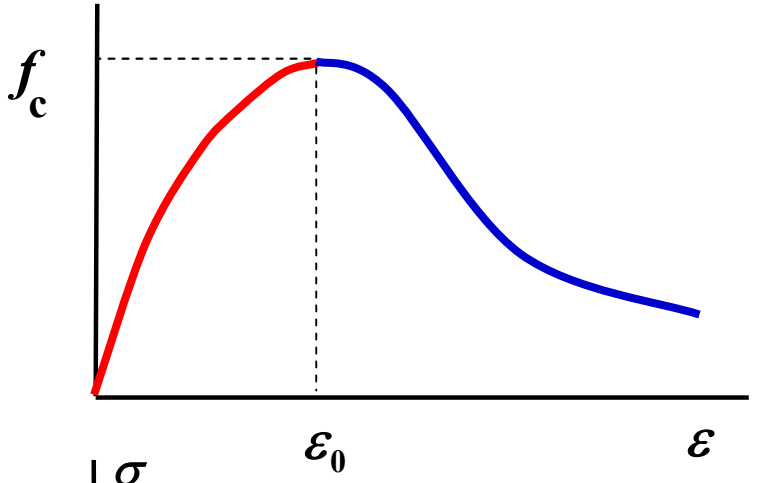
$\varepsilon_0 = ?$　0.002

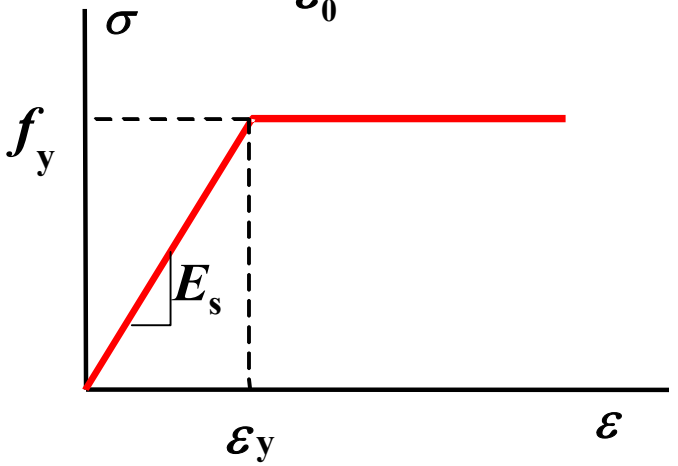
$\varepsilon_y = ?$

HRB335　0.00168

HRB400　0.002

HRB500　0.0025





$$N = \sigma_c A_c + \sigma_s A_s$$

2 轴心受压构件的受力性能Behavior of Axial Compressive Member

矩形截面轴心受压**短柱**

极限轴压承载力？

$\varepsilon_0 = ?$　　0.002

$\varepsilon_y = ?$

HRB335　　0.00168

HRB400　　0.002

HRB500　　0.0025

变形条件：$\varepsilon_s = \varepsilon_c = \varepsilon$

物理关系：$\sigma_s = E_s \varepsilon \qquad \varepsilon \le \varepsilon_y = \dfrac{f_y}{E_s}$

$\sigma_s = f_y \qquad \varepsilon > \varepsilon_y$

$$\sigma_c = f_c \left[\frac{2\varepsilon}{\varepsilon_0} - \left(\frac{\varepsilon}{\varepsilon_0} \right)^2 \right] \qquad 0 \le \varepsilon \le \varepsilon_0$$

平衡条件：

$$N = \sigma_c A_c + \sigma_s A_s$$

2 轴心受压构件的受力性能Behavior of Axial Compressive Member

矩形截面轴心受压短柱

极限轴压承载力？

如果以

$$f_c A_c = f'_y A'_s$$ 为条件

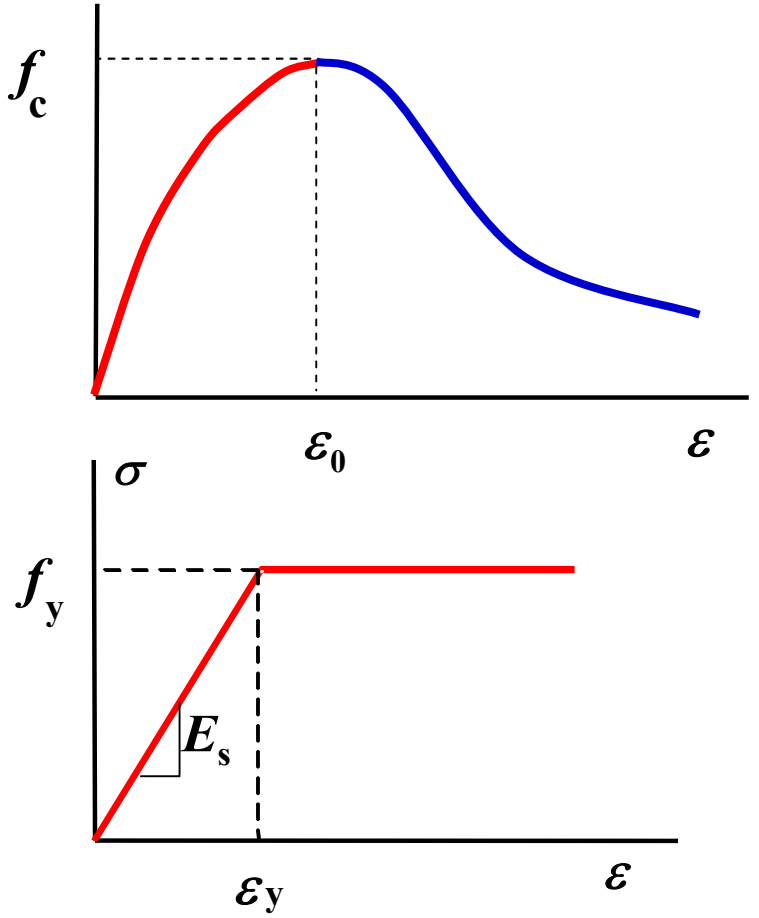
配筋率＝？

$$\rho' = \frac{A'_s}{A_c} = \frac{f_c}{f'_y}$$

20~50

300

$$\rho' = 6.7\% \sim 16.7\%$$



$$N = \sigma_c A_c + \sigma_s A_s$$

2 轴心受压构件的受力性能Behavior of Axial Compressive Member

矩形截面轴心受压**短柱**

极限轴压承载力？

如果以

$$f_c A_c = f'_y A'_s$$ 为条件

配筋率＝？

$$\rho' = \frac{A'_s}{A_c} = \frac{f_c}{f'_y}$$ **20~50** **300**

$$\rho' = 6.7\% \sim 16.7\%$$

变形条件：$\varepsilon_s = \varepsilon_c = \varepsilon$

物理关系：$\sigma_s = E_s \varepsilon$　　$\varepsilon \le \varepsilon_y = \dfrac{f_y}{E_s}$

$\sigma_s = f_y$　　$\varepsilon > \varepsilon_y$

$$\sigma_c = f_c \left[\frac{2\varepsilon}{\varepsilon_0} - \left(\frac{\varepsilon}{\varepsilon_0} \right)^2 \right] \qquad 0 \le \varepsilon \le \varepsilon_0$$

平衡条件：

$$N = \sigma_c A_c + \sigma_s A_s$$

2 轴心受压构件的受力性能Behavior of Axial Compressive Member

矩形截面轴心受压**短柱**

极限轴压承载力？

是以混凝土达到ε_0为承载力标志

达到极限承载力后，构件的破坏特征？

变形条件：$\varepsilon_s = \varepsilon_c = \varepsilon$

物理关系：$\sigma_s = E_s \varepsilon \qquad \varepsilon \le \varepsilon_y = \dfrac{f_y}{E_s}$

$$\sigma_s = f_y \qquad \varepsilon > \varepsilon_y$$

$$\sigma_c = f_c\left[\frac{2\varepsilon}{\varepsilon_0} - \left(\frac{\varepsilon}{\varepsilon_0}\right)^2\right] \qquad 0 \le \varepsilon \le \varepsilon_0$$

平衡条件：

$$N = \sigma_c A_c + \sigma_s A_s$$

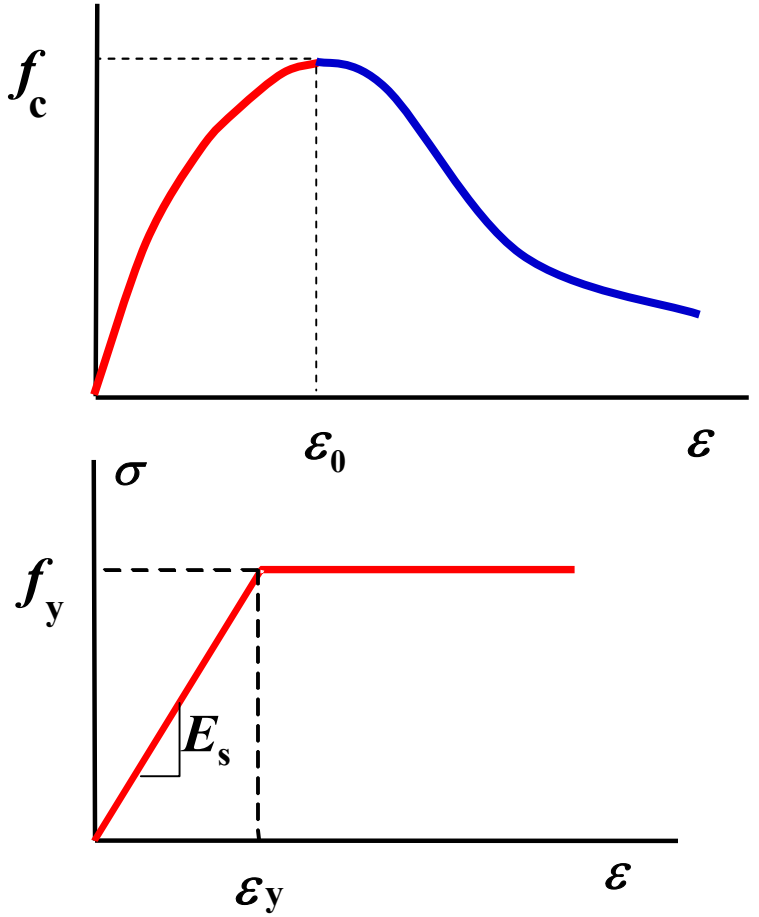
2 轴心受压构件的受力性能Behavior of Axial Compressive Member

矩形截面轴心受压短柱

极限轴压承载力？

是以混凝土达到ε_0为承载力标志

达到极限承载力后，构件的破坏特征？



$$N = \sigma_c A_c + \sigma_s A_s$$

$$N=\sigma_c A_c+\sigma_s A_s$$

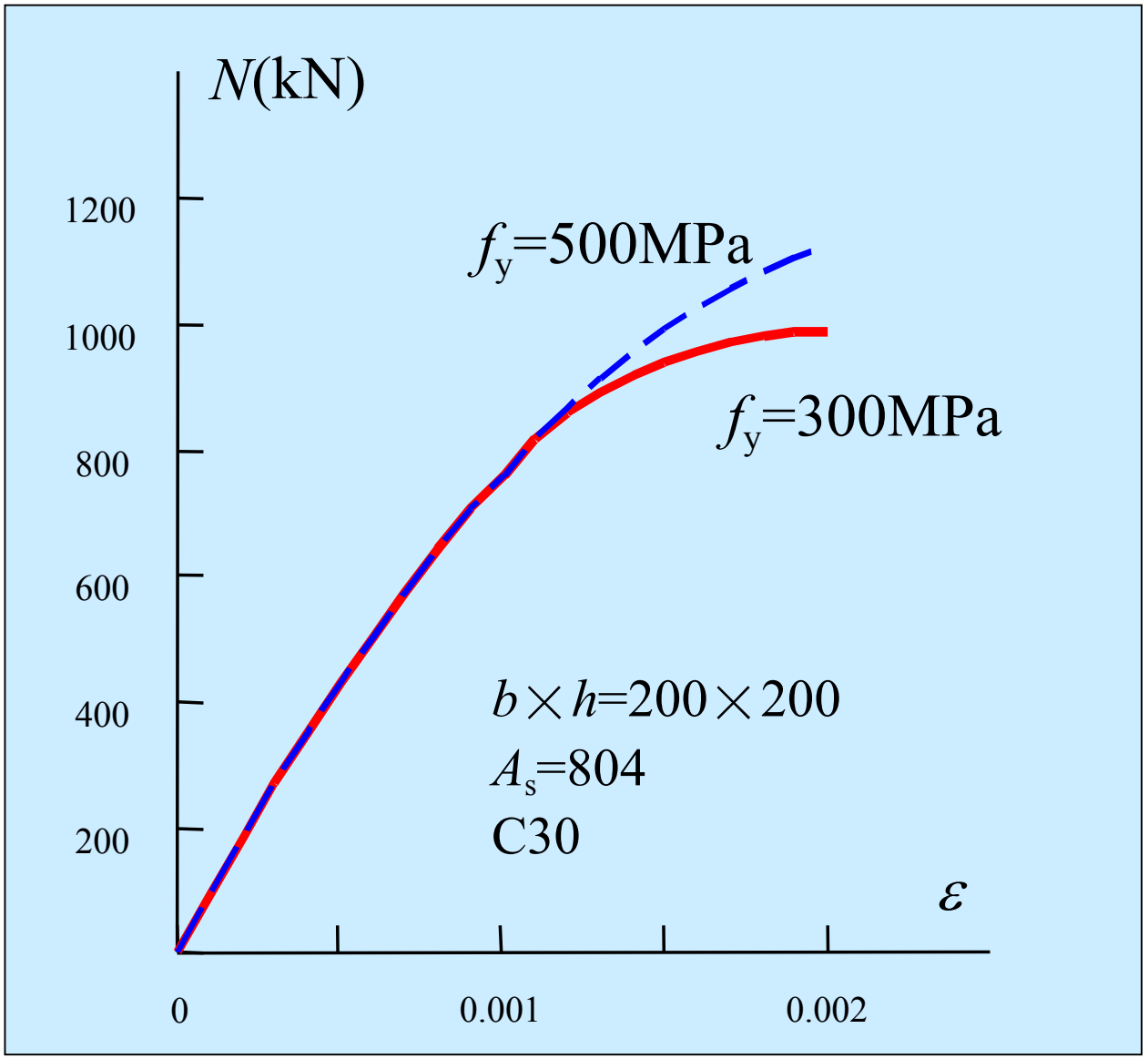
对于$\varepsilon_y=f_y/E_s<\varepsilon_0$的钢筋　　**对于$\varepsilon_y=f_y/E_s>\varepsilon_0$的钢筋**

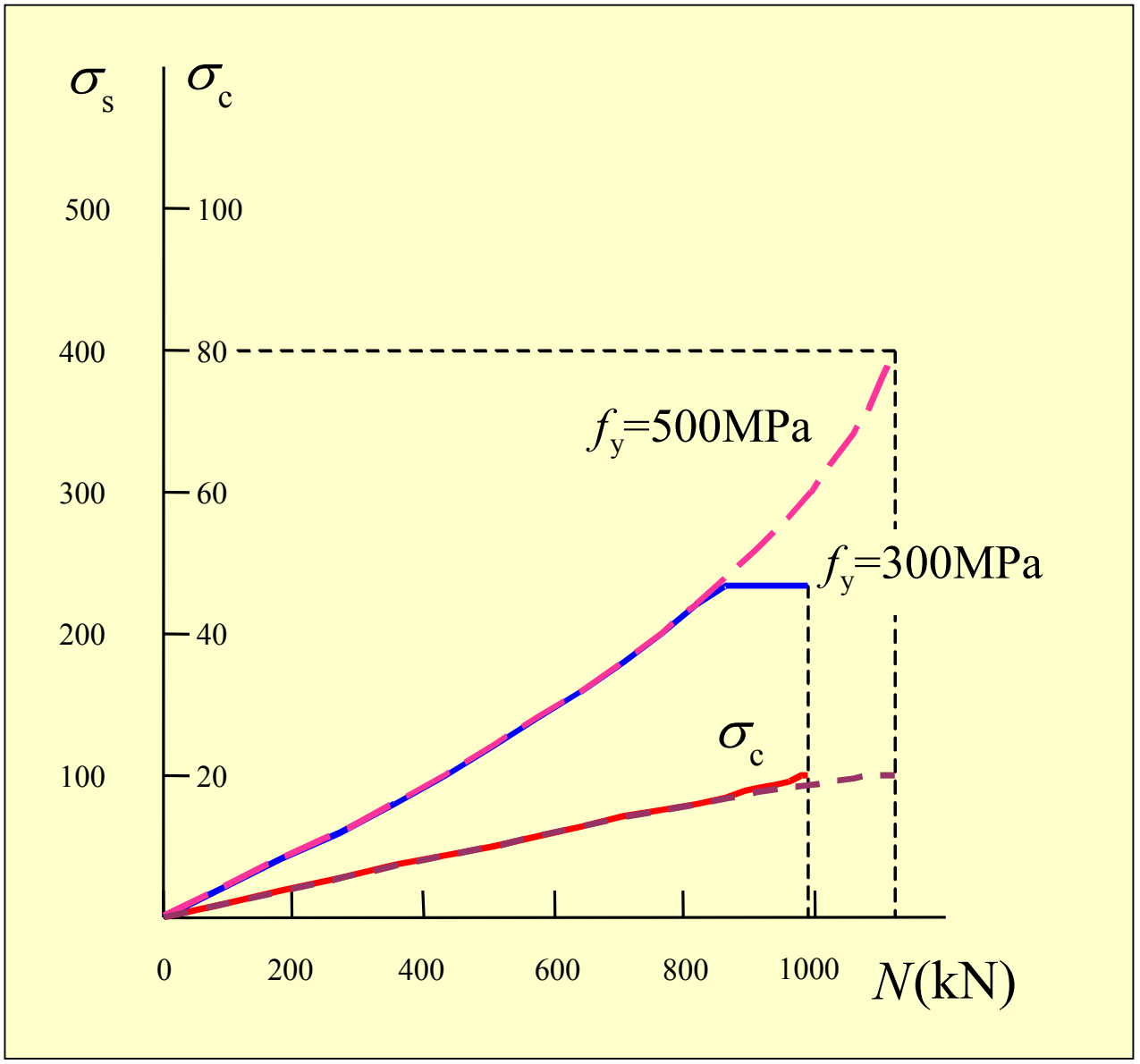
当$\varepsilon\le\varepsilon_y$时　$N=f_c\left[\dfrac{2\varepsilon}{\varepsilon_0}-\left(\dfrac{\varepsilon}{\varepsilon_0}\right)^2\right]A_c+E_s\varepsilon A_s$

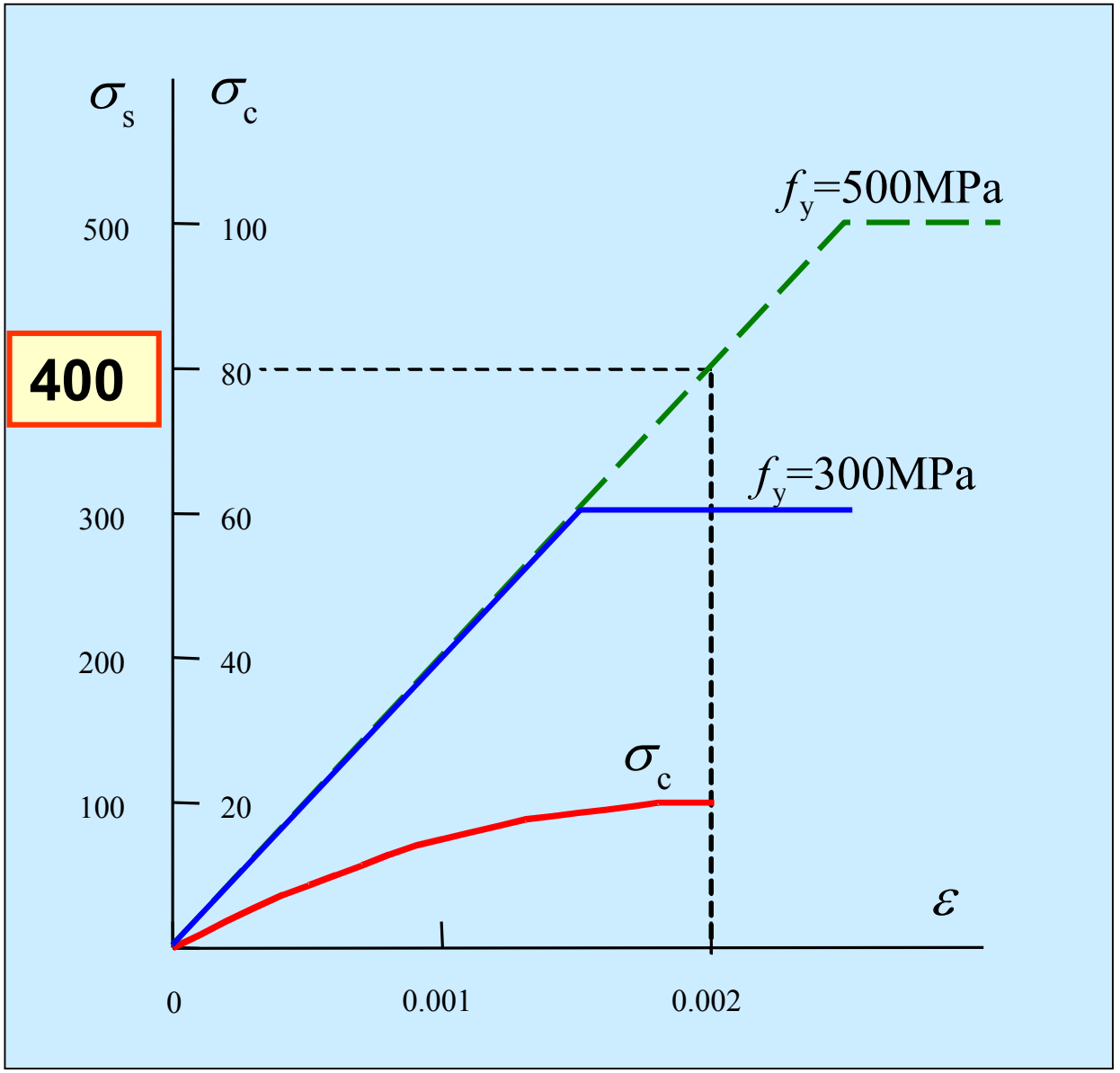
当$\varepsilon>\varepsilon_y$时　$N=f_c\left[\dfrac{2\varepsilon}{\varepsilon_0}-\left(\dfrac{\varepsilon}{\varepsilon_0}\right)^2\right]A_c+f_y A_s$

当$\varepsilon=\varepsilon_0$时　$N_u=f_c A_c+f_y A_s$　　$N_u=f_c A_c+E_s\varepsilon_0 A_s$

$E_s\varepsilon_0=2\times10^5\times0.002=400\text{MPa}$


N(kN)
1200
1000
800
600
400
200
f_y=500MPa
f_y=300MPa
$b \times h$=200×200
A_s=804
C30
ε
0
0.001
0.002


σs
σc
500
100
400
80
300
60
200
40
100
20
fy=500MPa
fy=300MPa
σc
0
200
400
600
800
1000
N(kN)

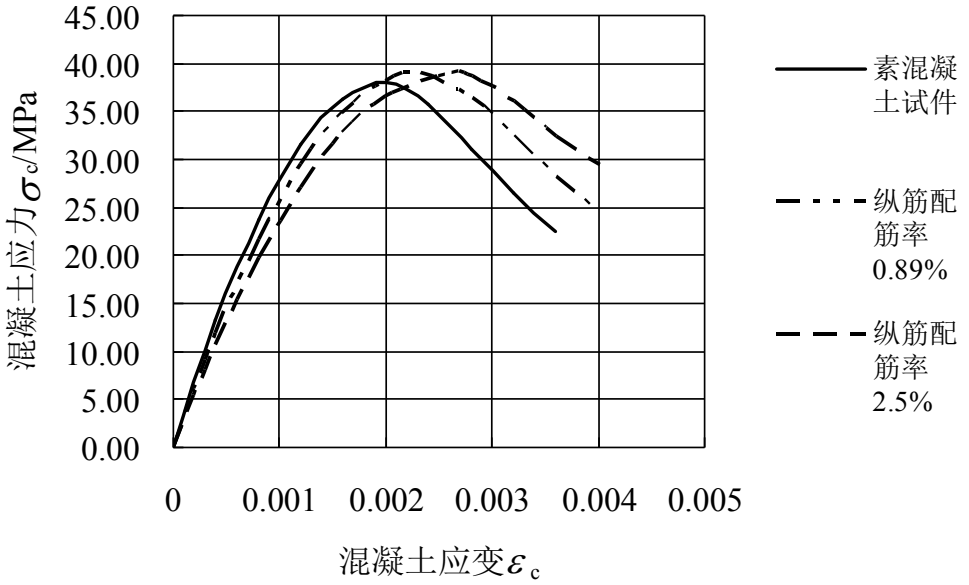

σs
σc
500
100
400
80
300
60
200
40
100
20
fy=500MPa
fy=300MPa
σc
ε
0
0.001
0.002

钢筋的受压强度

$$\begin{cases} \varepsilon_y = \dfrac{f_y}{E_s} < \varepsilon_0 \text{时}, \qquad f'_y = f_y \\ \varepsilon_y = \dfrac{f_y}{E_s} > \varepsilon_0 \text{时}, \qquad f'_y = E_s\varepsilon_0 = 400\text{MPa} \end{cases}$$

HRB500级钢筋的抗压强度=？





30根配有500MPa级纵向受力钢筋的轴心受压短柱(混凝土强度等级C30~C80，受压钢筋配筋率0.2%~5.2%)的试验结果表明，在配有纵筋的受压构件中，混凝土的受压的峰值应变有所提高；当轴心压力增大到破坏荷载的90%左右时，柱侧面混凝土出现纵向裂缝，此时混凝土的压应变已达到0.0025，纵向受压钢筋的应力已达到500MPa。

轴心受压承载力

$$N_u = f_c A_c + f'_y A_s$$

表 4.2.3-1　　普通钢筋强度设计值(N / mm2)

种　类		符号	f_y	f'_y
热轧钢筋	HPB 235(Q235)	Φ	210	210
	HRB 335(20MnSi)	Φ	300	300
	HRB 400(20MnSiV、20MnSiNb、20MnTi)	Φ	360	360
	400(K20MnSi)	ΦR	360	360

注：在钢筋混凝土结构中，轴心受拉和小偏心受拉构件的钢筋抗拉强度设计值大于 300N / mm^2 时，仍应按 300N/mm^2 取用。

新规范的变化

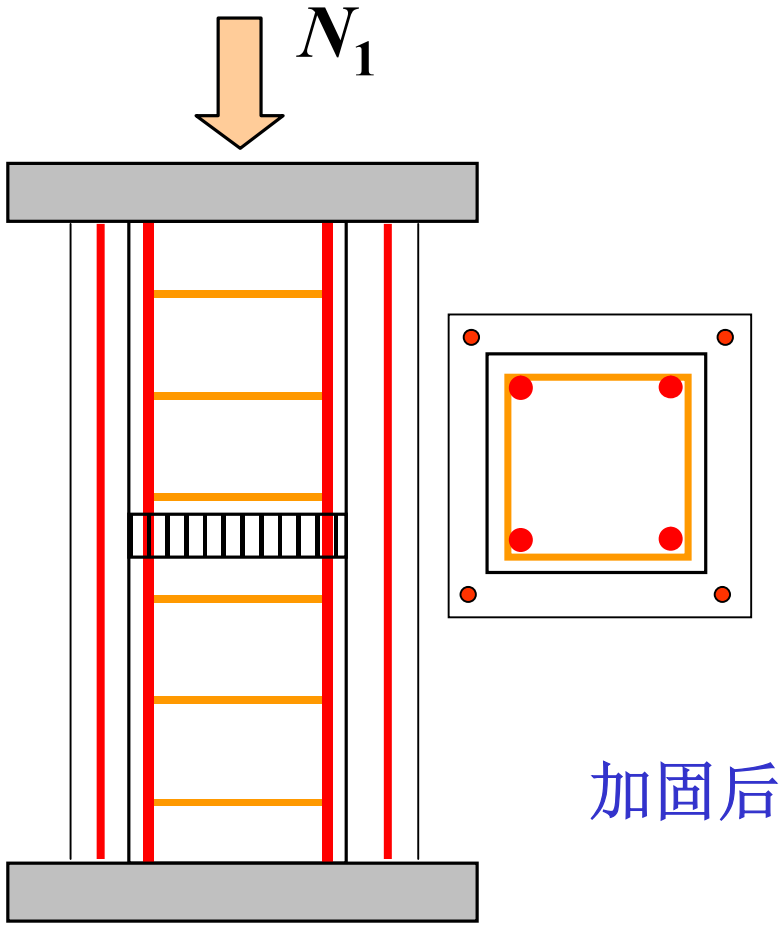
表 4.2.3-1　钢筋强度设计值（N/mm2）

种类	f_y	f'_y
HPB300	270	270
HRB335、HRBF335	300	300
HRB400、HRBF400、RRB400	360	360
HRB500、HRBF500、RRB500	435	435

注：1　用作受剪、受扭、受冲切承载力计算的箍筋，抗拉设计强度f_{yv}按表中f_y的数值取用，但其数值不应大于 360 N/mm^2；

2　用作局部承压的间接配筋，以及受压构件约束混凝土配置的箍筋，抗拉设计强度f_y按表中的数值取用。

思考题



加固后的极限承载力该怎么计算？

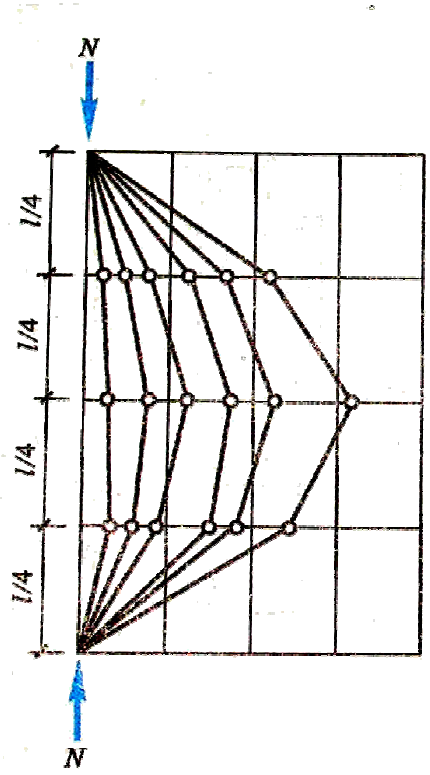
- 搞清基本受力过程
- 抓住关键受力状态
- 掌握三大法宝：变形协调、物理关系、受力平衡

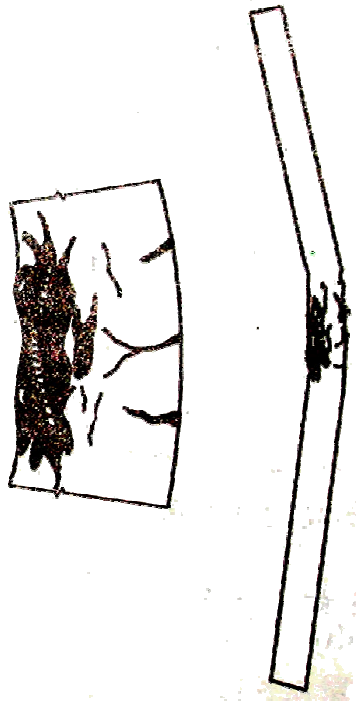
轴心受压承载力设计

$$N \le 0.9\varphi(f_c A_c + f'_y A_s)$$

稳定系数

$$N \le 0.9\varphi(f_c A_c + f'_y A_s)$$





轴心受压承载力设计

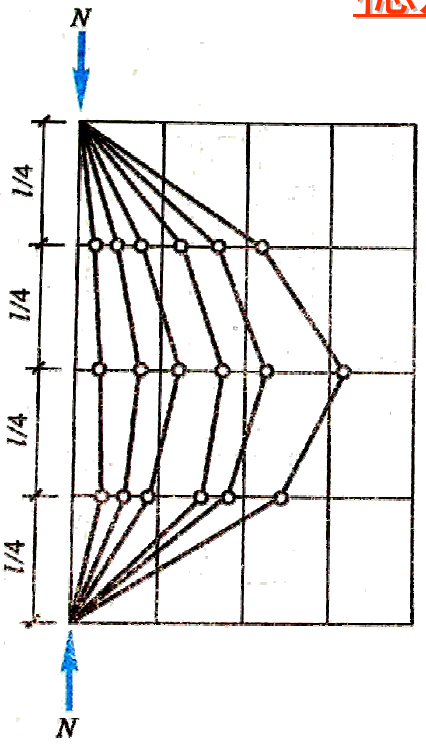
$$N \le 0.9\varphi(f_c A_c + f'_y A_s)$$

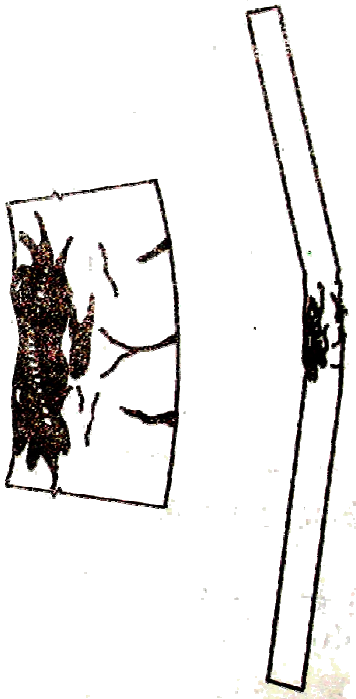
折减系数 0.9是考虑初始偏心(accidental eccentricity)的影响，以及主要承受恒载作用的轴压受压柱的可靠性。

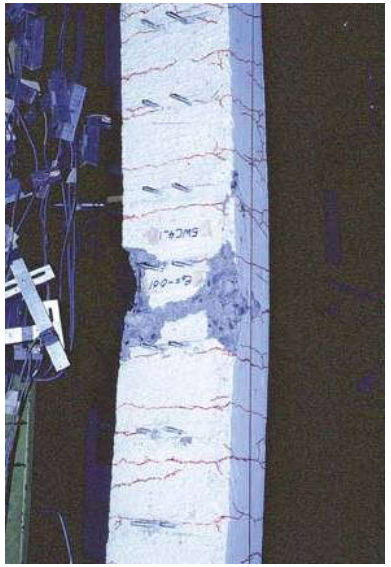
轴心受压承载力设计

$$N \le 0.9\varphi(f_c A_c + f'_y A_s')$$

稳定系数







轴心受压承载力设计

$$N \le 0.9\varphi(f_c A_c + f'_y A_s)$$

稳定系数

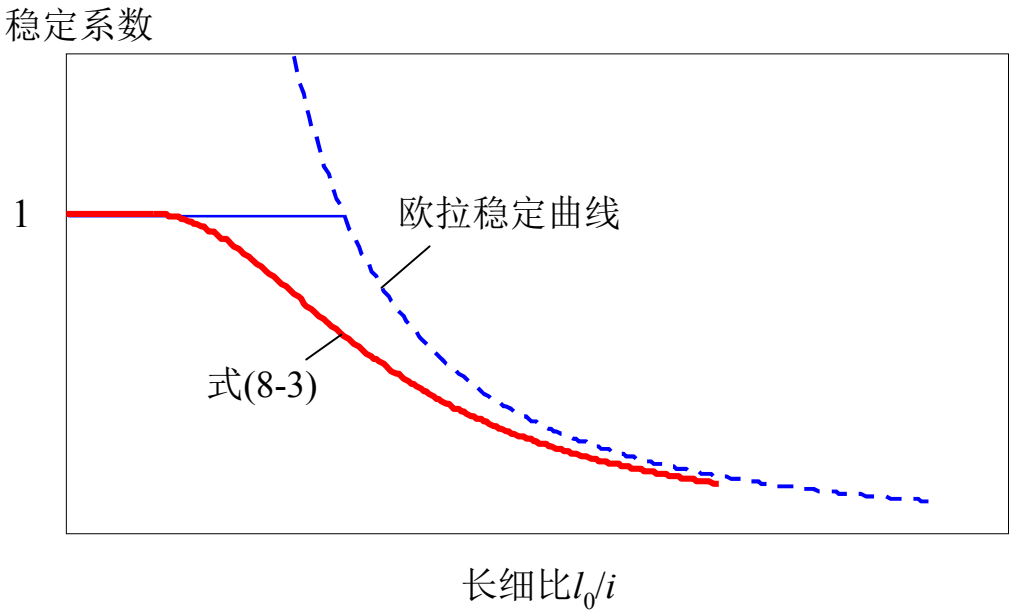
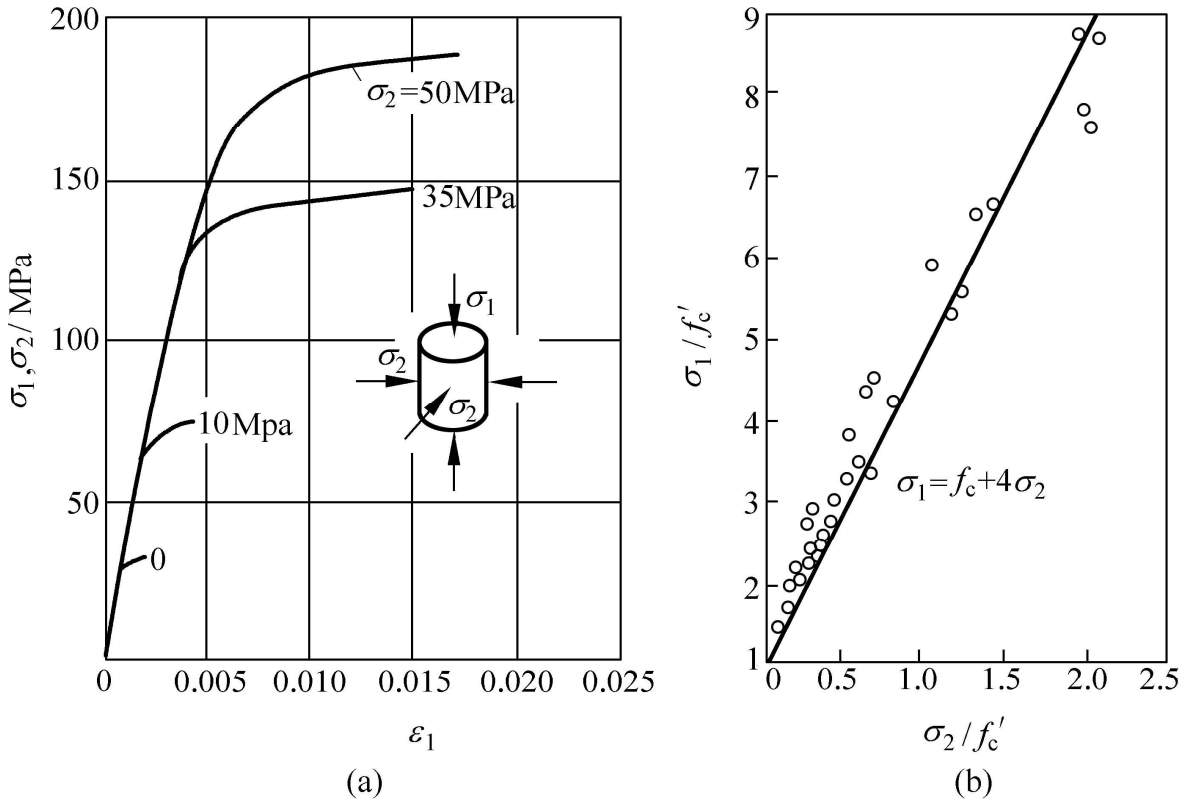


表 8-1　钢筋混凝土轴心受压构件的稳定系数φ

l_0/b	≤8	10	12	14	16	18	20	22	24	26	28
l_0/d	≤7	8.5	10.5	12	14	15.5	17	19	21	22.5	24
l_0/i	≤28	35	42	48	55	62	69	76	83	90	97
φ	1.0	0.98	0.95	0.92	0.87	0.81	0.75	0.70	0.65	0.60	0.56
l_0/b	30	32	34	36	38	40	42	44	46	48	50
l_0/d	26	28	29.5	31	33	34.5	26.5	38	40	41.5	43
l_0/i	104	111	118	125	132	139	146	153	160	167	174
φ	0.52	0.48	0.44	0.40	0.36	0.32	029	0.26	0.23	0.21	0.19

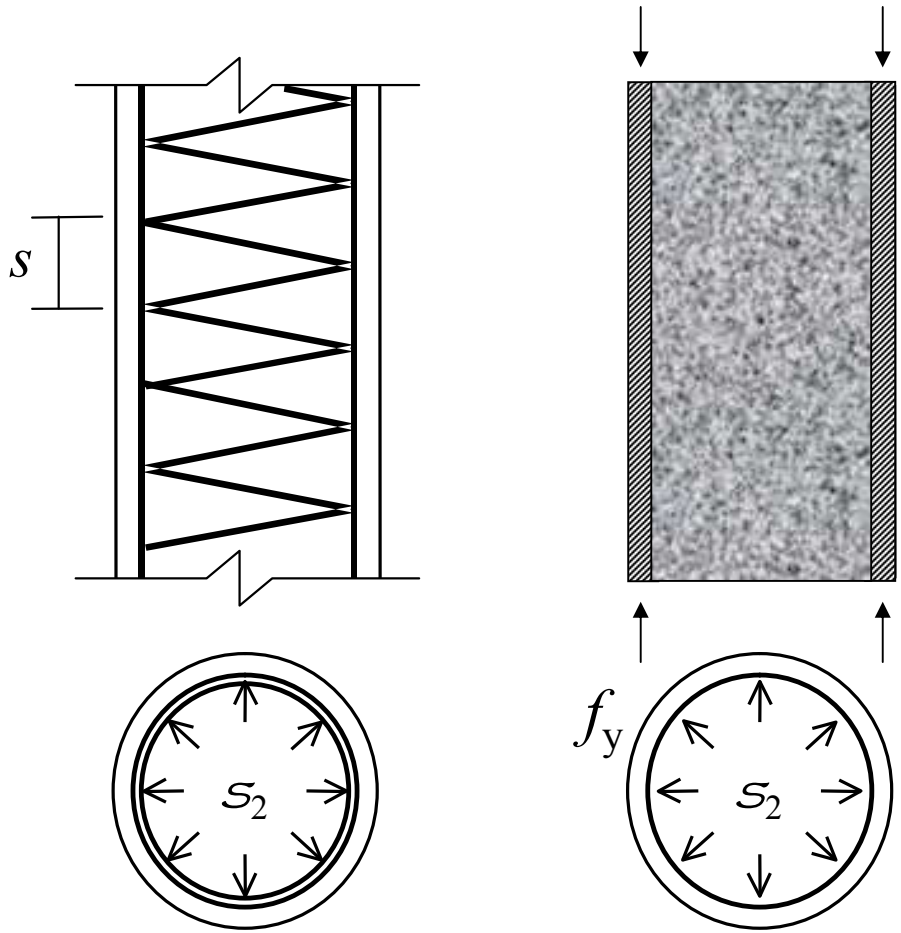
约束混凝土

$$f'_{c1} = f'_c + 4\sigma_2$$

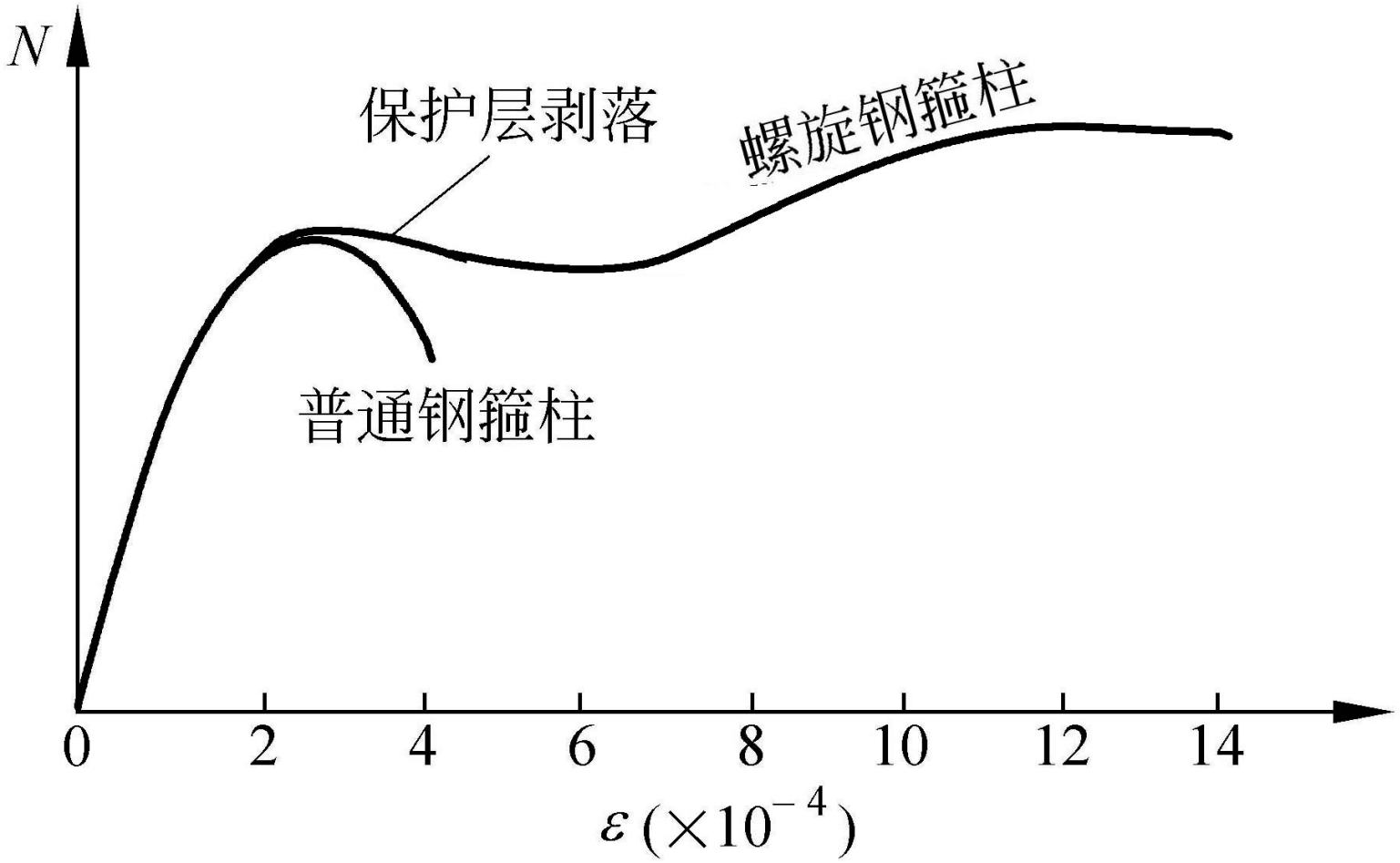


约束混凝土柱

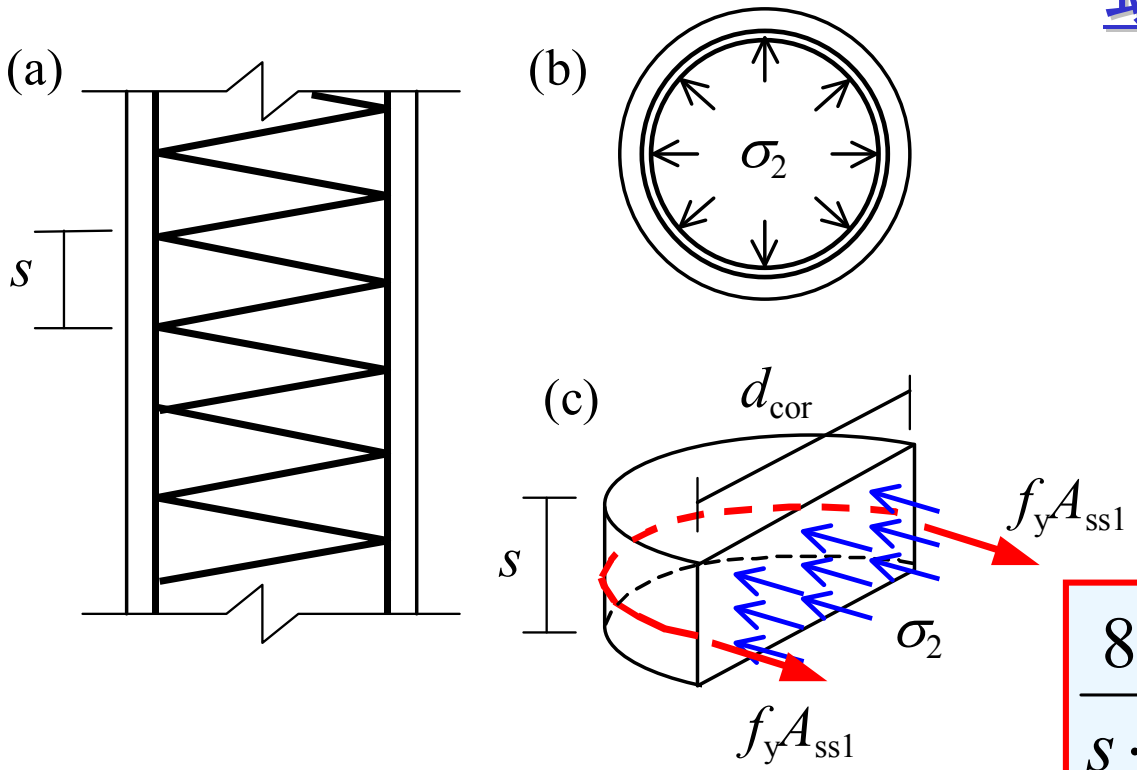
螺旋箍筋柱　　钢管混凝土柱　　FRP约束混凝土柱



螺旋箍筋混凝土柱



螺旋箍筋混凝土柱



$$\pi d_{cor} A_{ss1} = s \cdot A_{ss0}$$

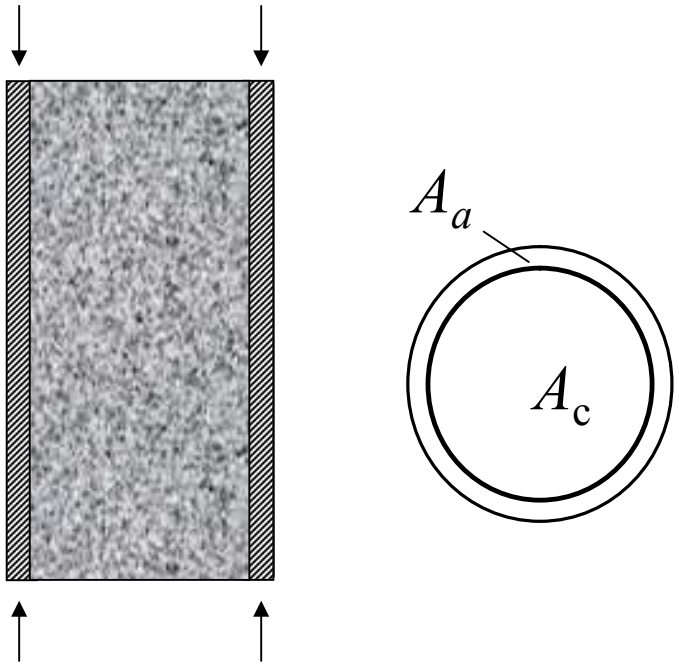
$$A_{ss0} = \frac{\pi d_{cor} A_{ss1}}{s}$$

$$\frac{8A_{ss1}}{s \cdot d_{cor}} \cdot \frac{\pi d_{cor}^2}{4} = 2\frac{\pi d_{cor} A_{ss1}}{s}$$

$$N \le N_u = 0.9(f_c A_{cor} + f'_y A'_s + 2\alpha f_y A_{ss0})$$

螺旋箍筋对承载力的影响系数α，当$f_{cu,k} \le 50N/mm^2$时，取$\alpha = 1.0$；当$f_{cu,k} = 80N/mm^2$时，取$\alpha = 0.85$，其间直线插值。

钢管混凝土柱



简化计算公式

$$N \le N_u = f_c A_c (1 + 2\sqrt{\theta})$$

$$\theta = \frac{f_y A_a}{f_c A_c}$$

钢管和混凝土都处于三向受力状态

轴心受拉性能

Behavior of uniaxial tensile RC member

轴心受压性能

Behavior of uniaxial compressive RC member

- 搞清基本受力过程
- 抓住关键受力状态
- 掌握三大法宝：变形协调、物理关系、受力平衡
- 熟悉设计公式

- 工程设计是一个在多种约束条件下寻找合理“解”的过程。

- 所谓约束条件，是指工程建设的用途、规模、投资、业主要求、环境、地理、材料供应、安全、施工技术水平等，这些条件应尽量满足。

- 工程管理是对获得合理“解”过程的控制。

需要获得什么样的“解”？

?+?=10　　3+7=?

设计 ≠ 计算

约束条件 → 合理的方案

计算方法 → 准确的结果

综合和比较　　分析与推理

n个方程 **n+**个未知数　　**n**个方程 **n**个未知数

额外约束条件，合理假设

3.0 引言

Introduction

3.1 轴心受拉构件的受力性能



Behavior of Uniaxial Tensile Member

3.2 轴心受拉构件的设计

Design on Uniaxial Tensile Member

3.3 轴心受压构件的受力性能

Behavior of Uniaxial Compressive Member

3.4 轴心受压构件的设计

Design on Uniaxial Compressive Member

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Behavior of Uniaxial Tensile Member

3.2 轴心受拉构件的设计

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